agenda/afternoon **– breakout rooms**

Session 1: connecting multimodal data to brain cell types

Room 1255-1265

1:10-1:50 | defining and using common coordinate frameworks (reference atlases, mesoscale)

1:50-2:00 | patch-seq overview and tools for analysis

2:00-2:20 | Troubleshooting Annotation with the BKP

2:20-2:30 | break

2:30-2:40 | ABC Atlas demo & external use case examples

2:40-3:40 | user exploration of allen institute tools

3:40-4:00 | feedback survey

4:00 – 5:00 | additional time for user exploration

github.com/AllenInstitute/SPOCTT/tree/main/2026_Additional_Presentations

defining and using common coordinate frameworks/

Ashwin Bhandiwad, Scientist II

allen institute/

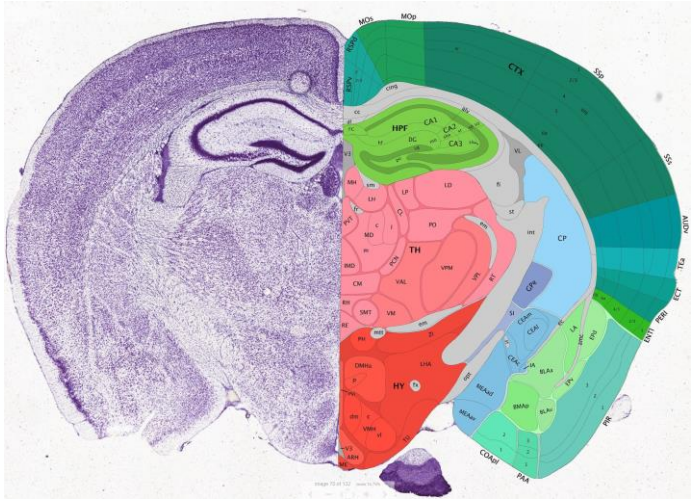
why do atlases matter?

- Common coordinate frameworks are the foundation of **spatial data integration**

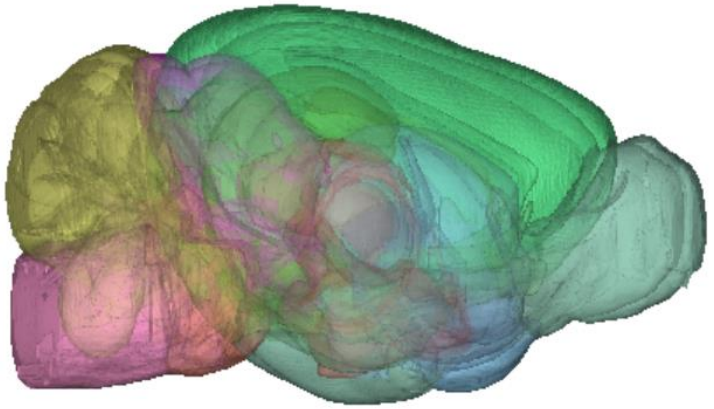
Use cases:

1. Reference for understanding how the brain is structured
2. Find datasets relevant to me (in or related to “my” brain region)
3. Compare and contrast different datasets / modalities

ARA (2008)



CCFv3 (2020)



atlas use cases/

Atlas data package

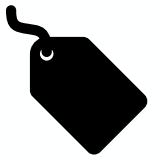


Vis tools



Understand how the brain is structured

Scientists tag samples



Semantic Search



Find datasets related to “my” brain region

Registration models/tools



Spatial search



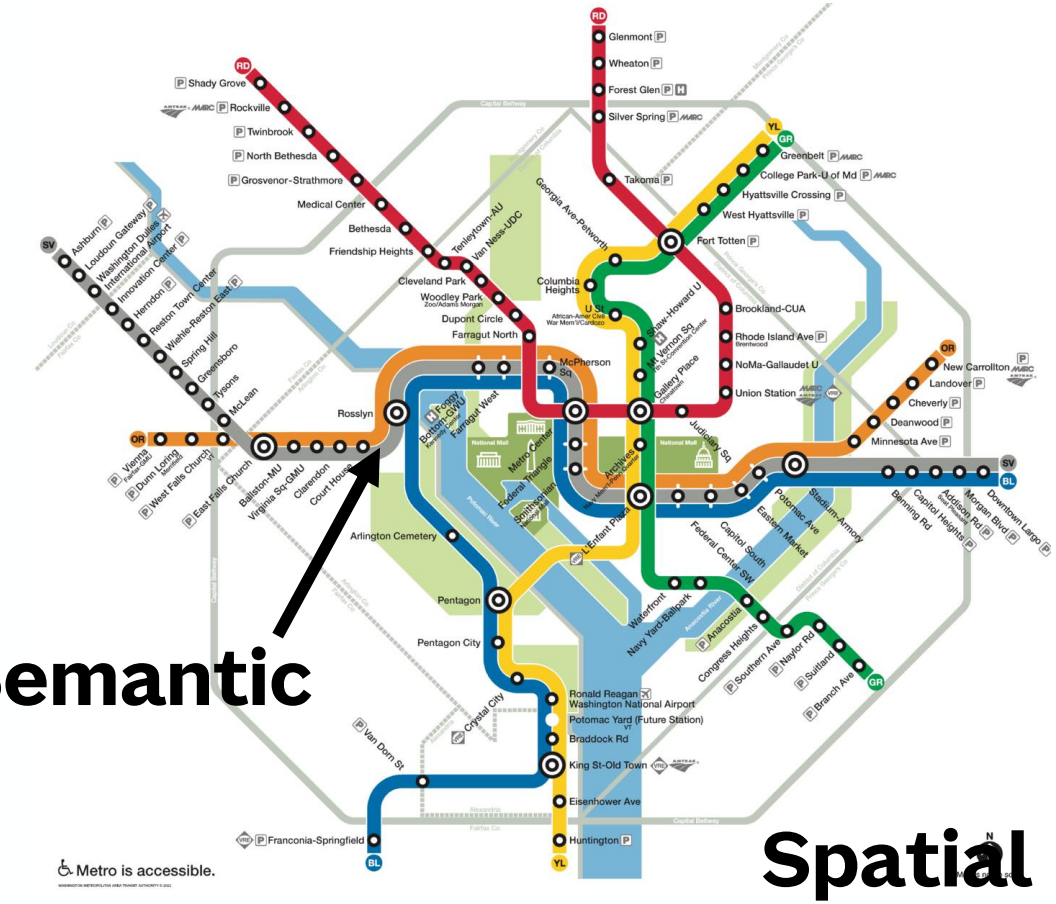
Compare and contrast datasets / modalities

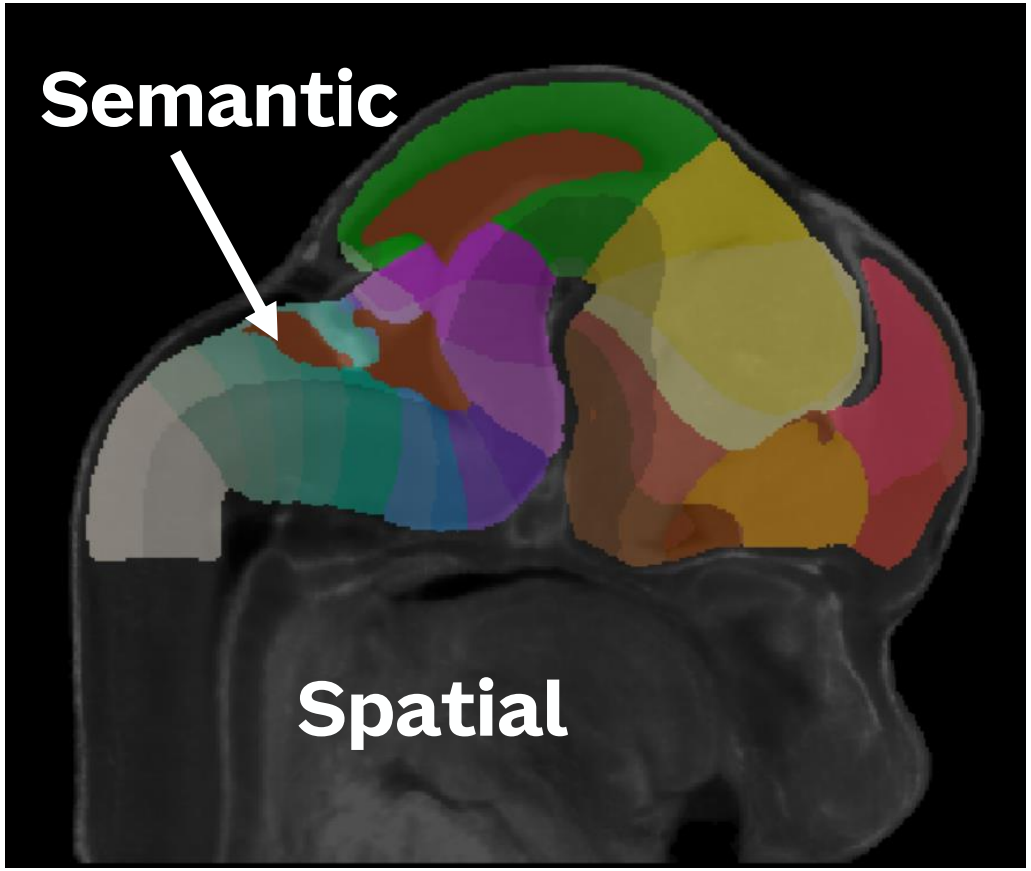
You are here



Higher complexity, higher impact

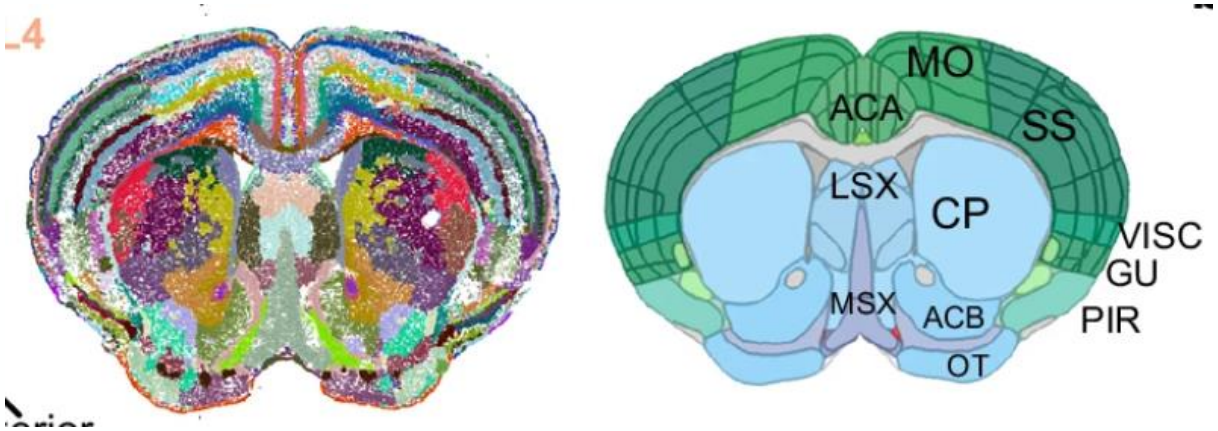
atlases have spatial and semantic components /





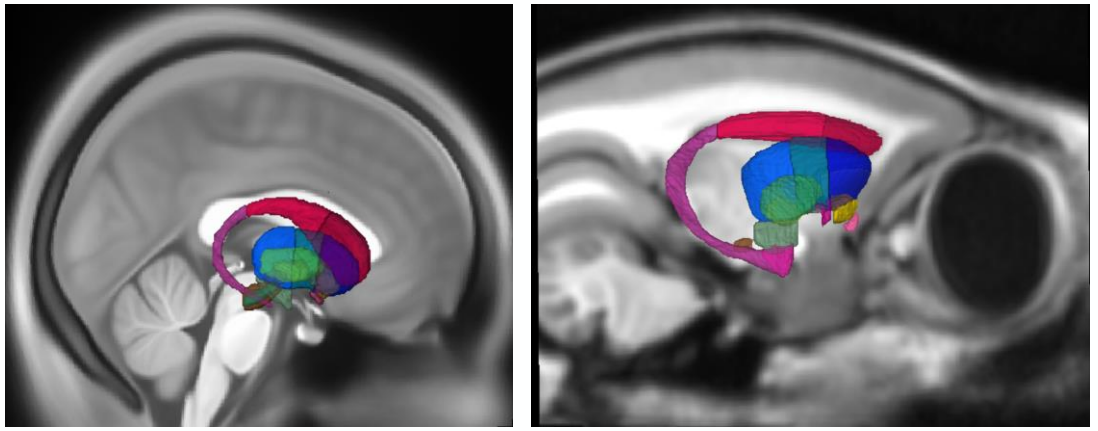
spatial and semantic layers can be changed/

Same ***space***
Different ***semantic***



CellTransformer
Lee et al. 2025

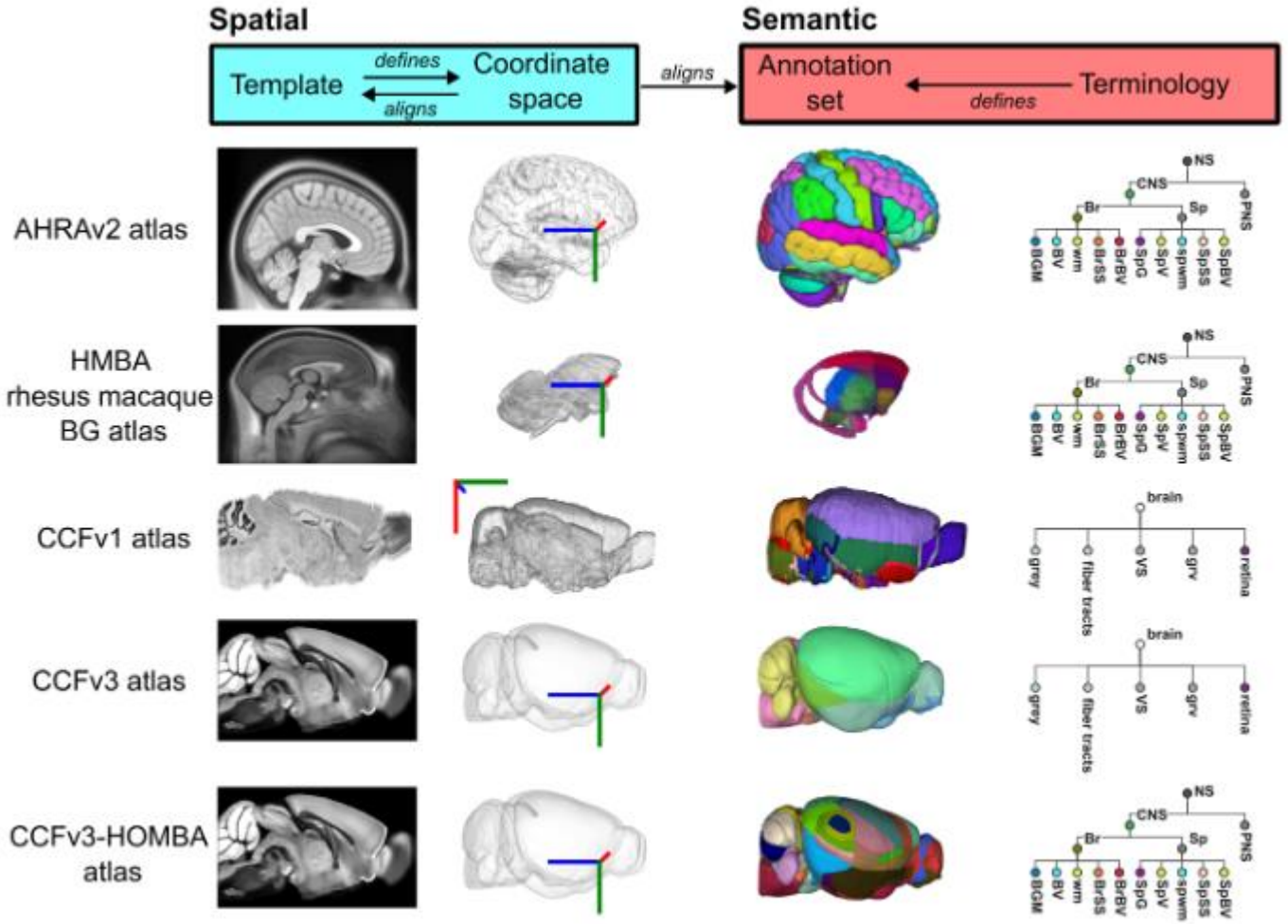
Different ***space***
same ***semantic***



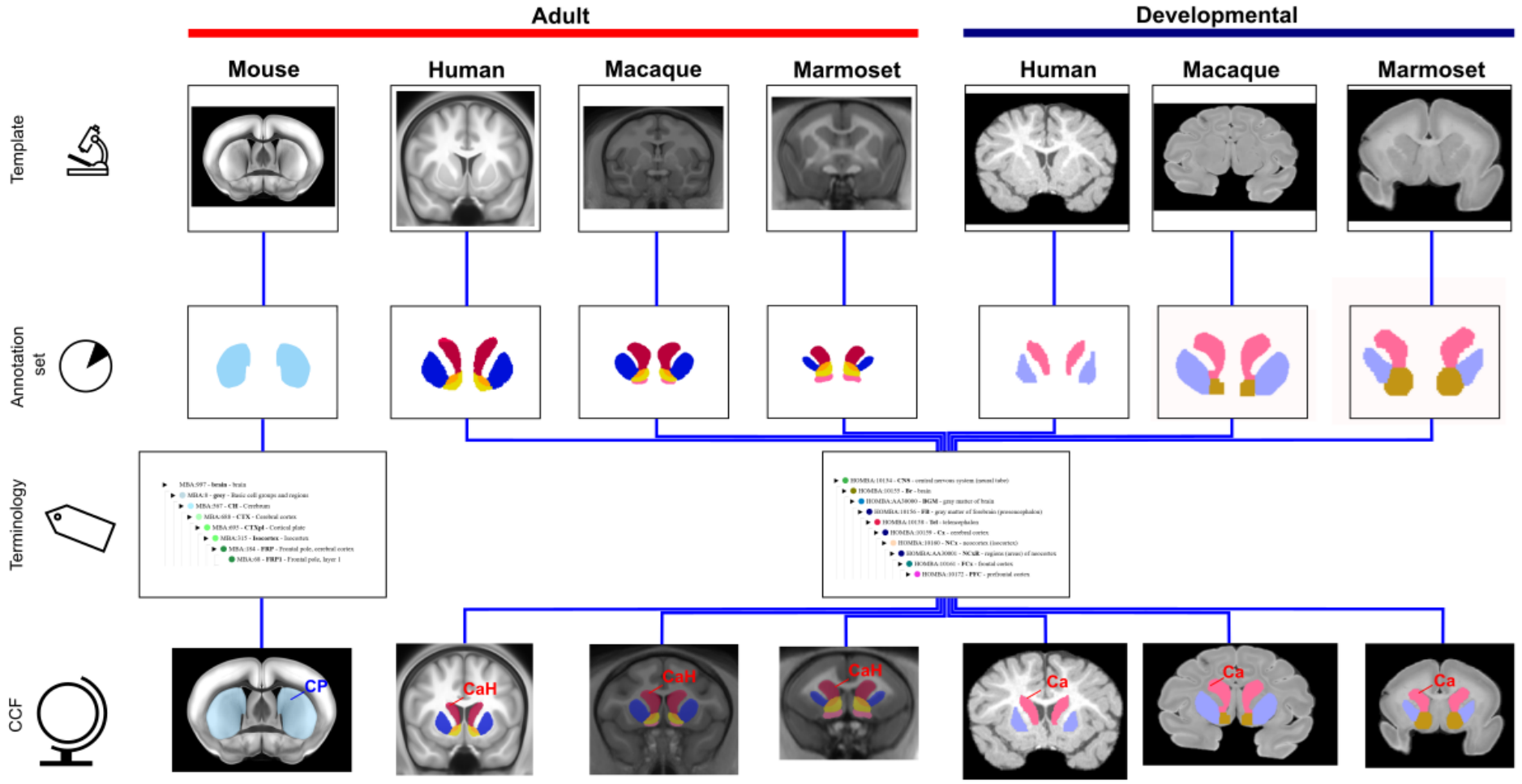
HMBA basal ganglia atlas
Ding et al. 2026

atlases are defined using a modular framework/

- Based on AtOM model (Kleven et al 2023)
- Every atlas must have at least ***one***:
  - Template
  - Coordinate space
  - Annotation set
  - Terminology
- Each individual component can be developed and versioned independent of the others
- **atlas assets** schema for hosting and versioning atlas components



terminology as an organizing principle/

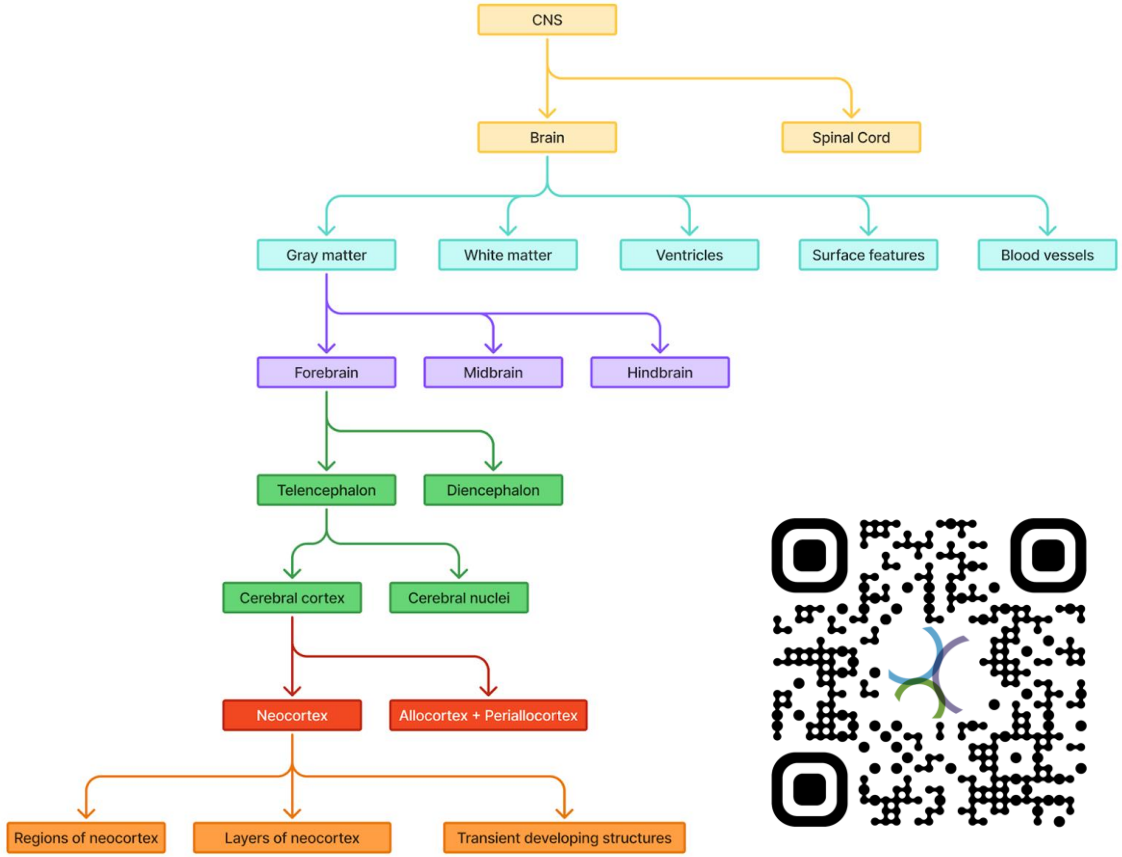


The BRAIN Initiative®

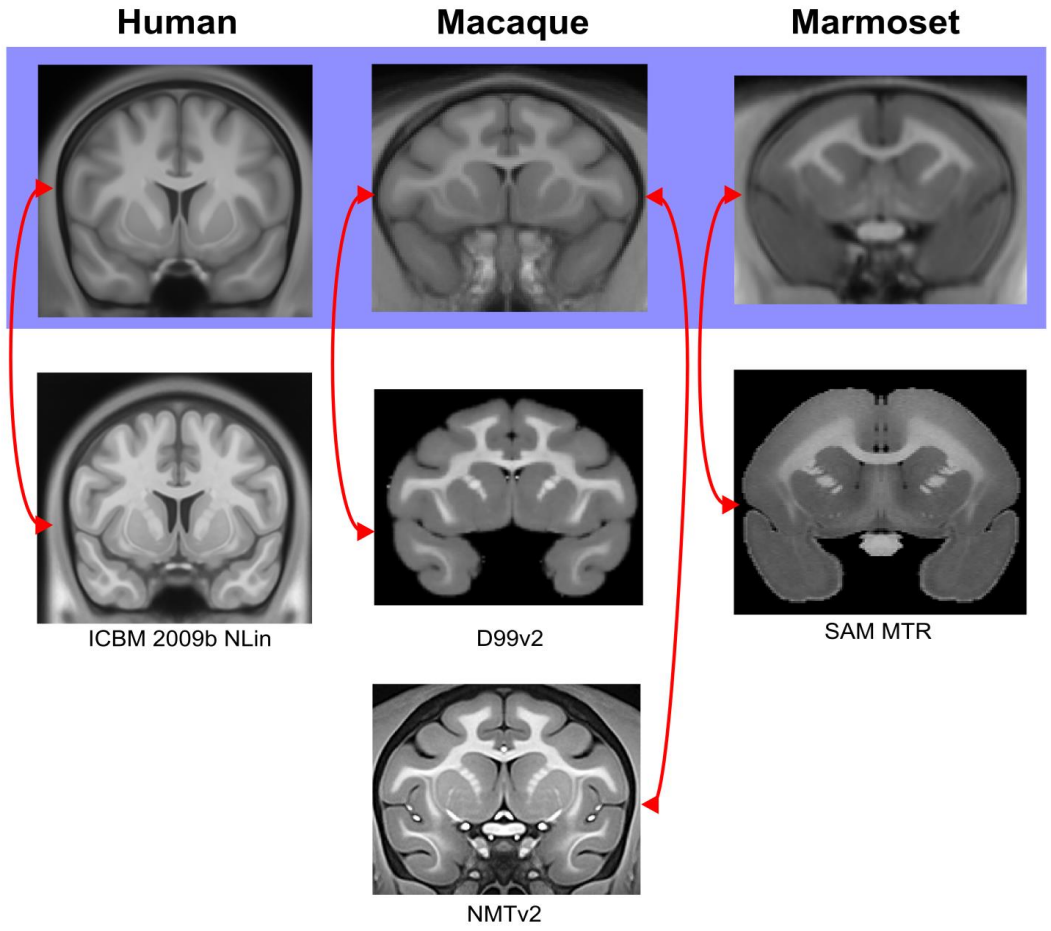
terminology as an organizing principle/

Harmonized ontology of mammalian brain anatomy (HOMBA)

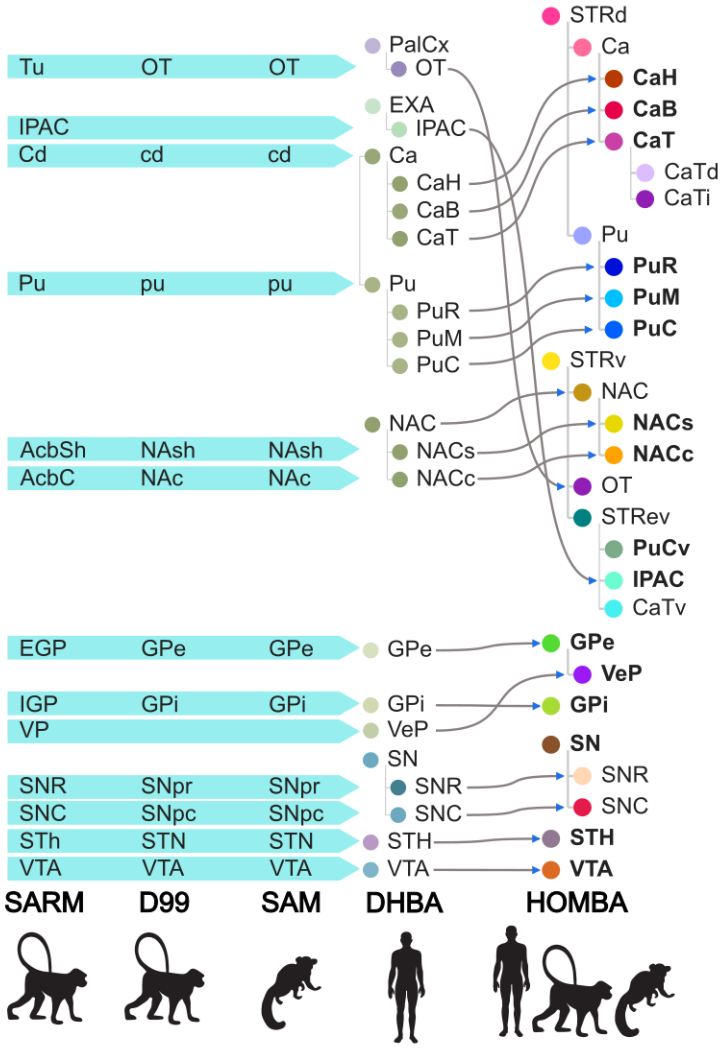
- HOMBA:10154 - **CNS** - central nervous system (neural tube)
  - HOMBA:10155 - **Br** - brain
    - HOMBA:AA30000 - **BGM** - gray matter of brain
      - HOMBA:10156 - **F** - gray matter of forebrain (prosencephalon)
        - HOMBA:10158 - **Tel** - telencephalon
          - HOMBA:10159 - **Cx** - cerebral cortex
          - HOMBA:10331 - **CN** - cerebral nuclei
            - HOMBA:10332 - **BN** - basal nuclei (basal ganglia)
              - HOMBA:AA30190 - **BNR** - regions of basal nuclei
                - HOMBA:10333 - **STR** - striatum (neostriatum)
                  - HOMBA:AA30191 - **STRre** - regions of striatum
                    - HOMBA:AA30192 - **STRd** - dorsal striatum (caudoputamen complex-CP)
                      - HOMBA:10334 - **Ca** - caudate nucleus (mediodorsal division of the CP)
                        - HOMBA:10335 - **CaH** - head of caudate nucleus
                          - HOMBA:11539 - **CaHld** - laterodorsal subdivision of CaH
                          - HOMBA:11541 - **CaHmv** - medioventral subdivision of CaH
                        - HOMBA:10336 - **CaB** - body of caudate nucleus
                          - HOMBA:AA30848 - **CaBld** - laterodorsal subdivision of CaB
                          - HOMBA:AA30849 - **CaBmv** - medioventral subdivision of CaB
                        - HOMBA:10337 - **CaT** - tail of caudate nucleus (caudolateral division of the CP)

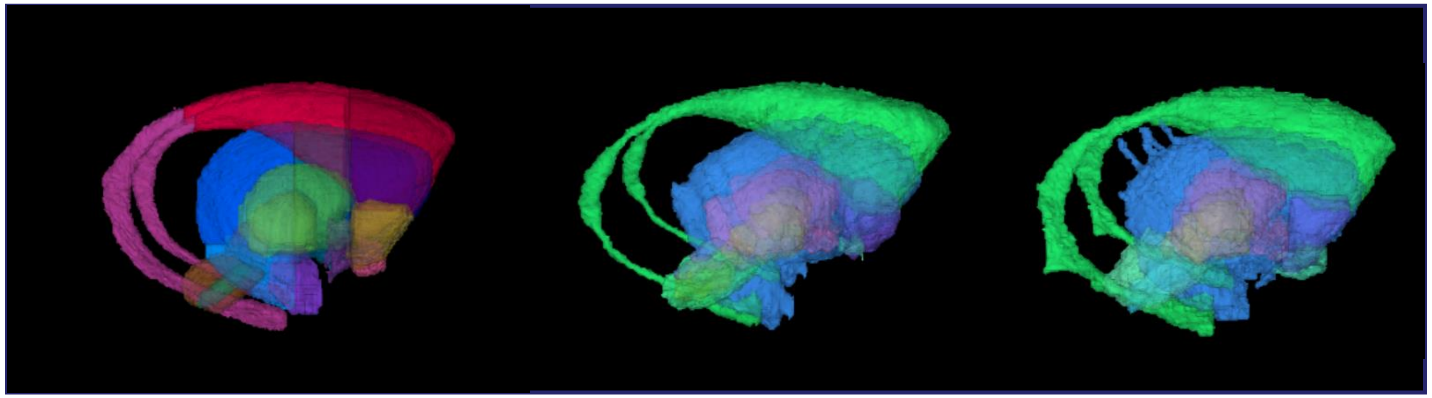


modularity unlocks interoperability/spatial layer



modularity unlocks interoperability/semantic layer





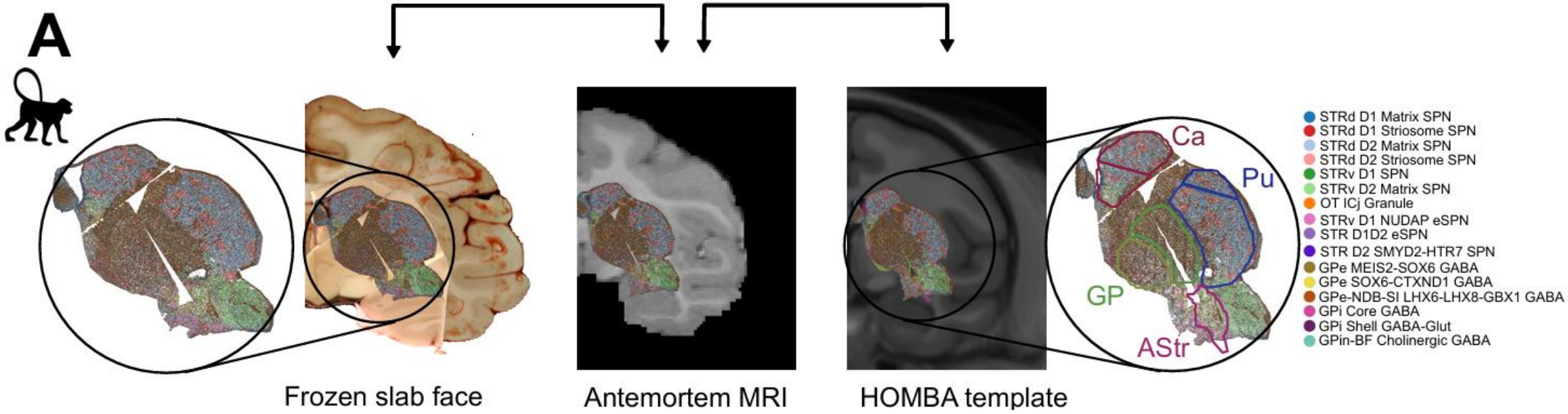
HOMBA (Ding et al. 2026) | D99v2 (Saleem et al. 2021) | SARM (Hartig et al. 2021)

how can I use this/examples



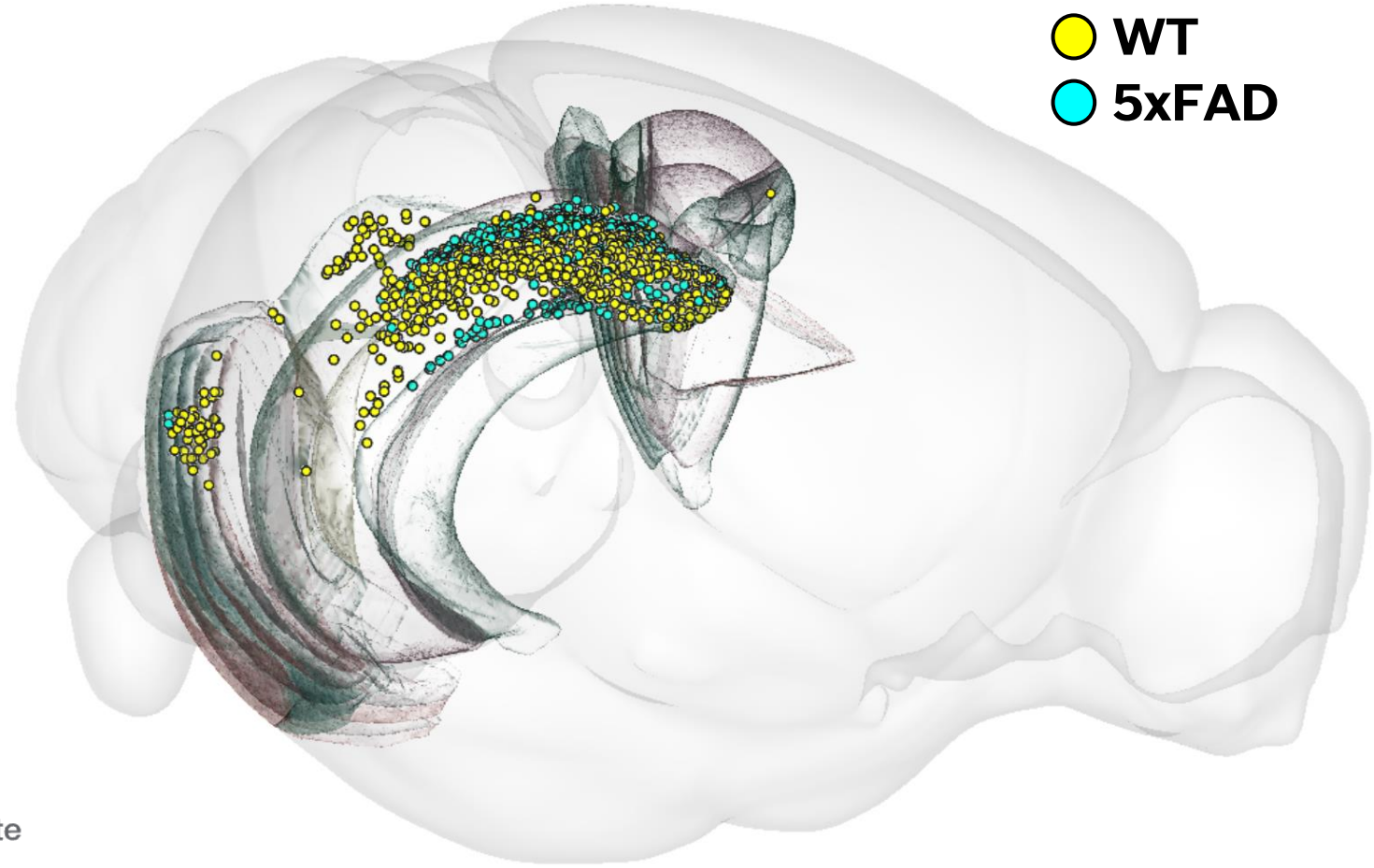
alleninstitute/

anatomical labels for spatial transcriptomic data/



Ding et al. 2026

topography of presynaptic connectivity loss/



NIH National Institute on Aging

Bhandiwad et al. 2026

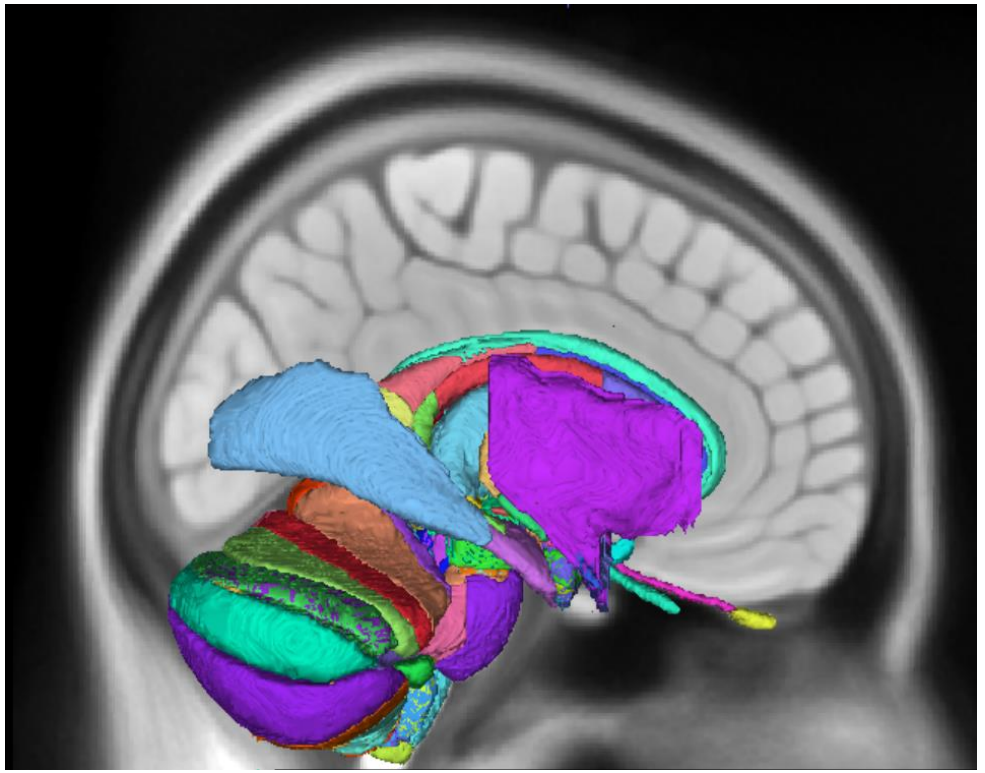
how can I use this/access



alleninstitute/

atlas access/manual access

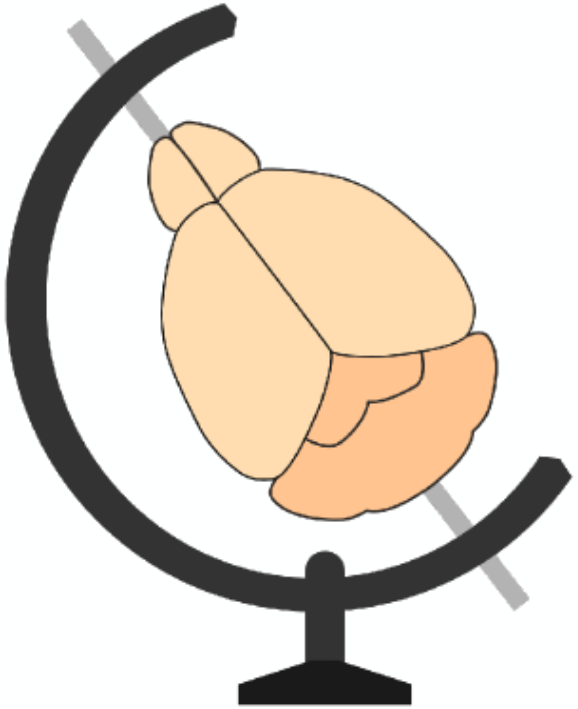
- All atlas data is available hosted on a public S3 bucket
  - Template and annotation set data are available in NIFTI or OME-Zarr formats
  - Terminology data are available in csv format
  - Coming soon: registration transforms between coordinate spaces
- Documentation and direct download links at the Atlas Primer
  - Links to visualization, interactive terminologies



atlas access/programmatic access

BrainGlobe

- All atlases use the atlas assets schema with assets in S3
  - Currently supports Allen mouse CCFv3 but launching HMBA human and NHP atlases by end of 2026
- Capability to update with new versions
- Common use cases:
  - Annotating coordinates with structural nomenclature
  - Creating structural/regional masks
  - Querying parent/child entities from an ontology
  - Labeling hemisphere



demo/

Google Colab link



allen institute/


alleninstitute/
thank you/

agenda/afternoon **– breakout rooms**

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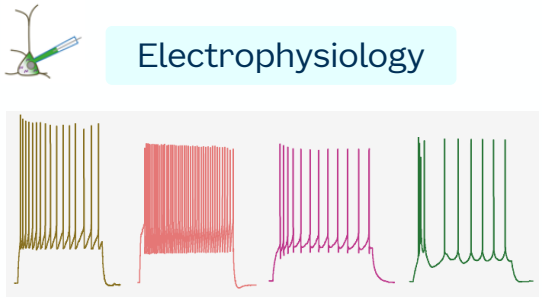
patch-seq overview and tools for analysis/

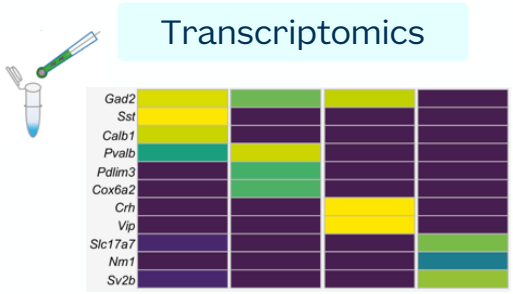
Rachel Hostetler, Scientist I

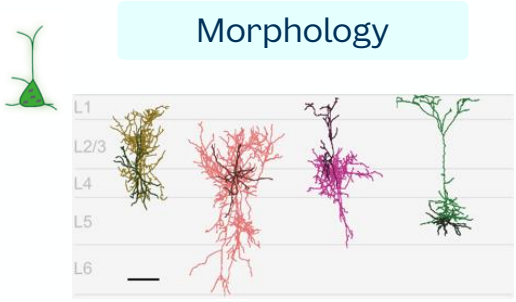
allen institute/

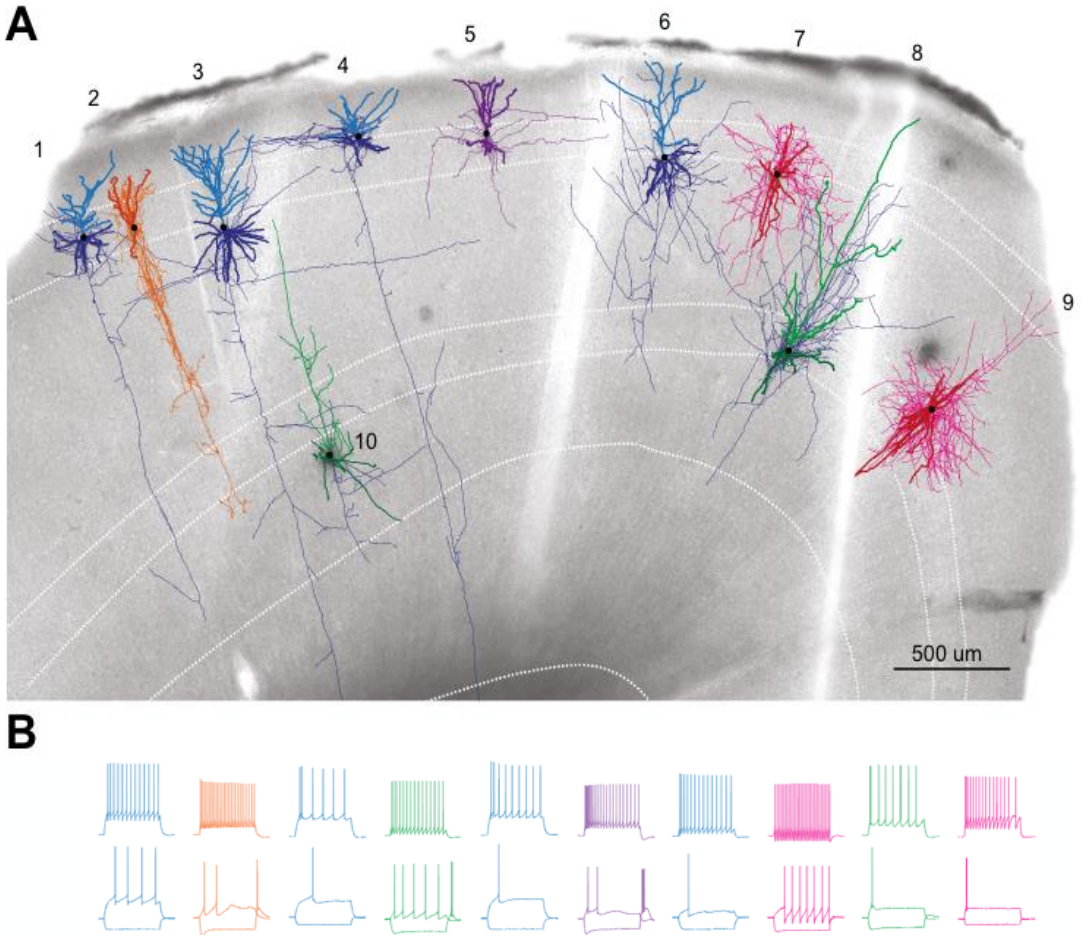
allen institute

patch-seq/ relate morpho-electric diversity to transcriptomic diversity







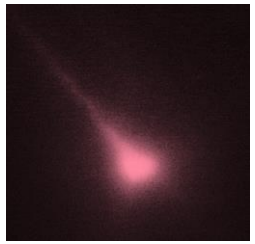


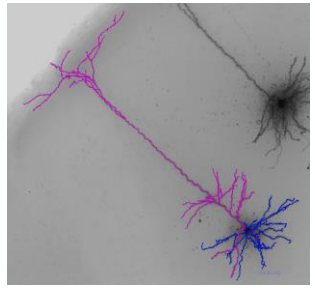
link cell shape and firing properties to cells types defined by gene expression

Lee, Budzillo, et al. (2021) eLife

patch-seq steps

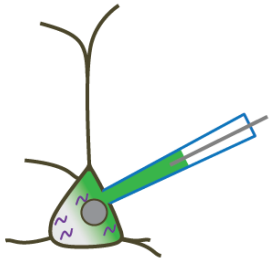
patch-seq in real life

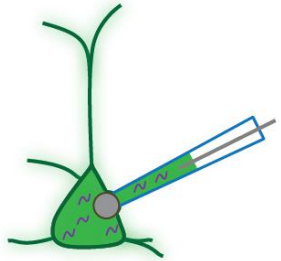


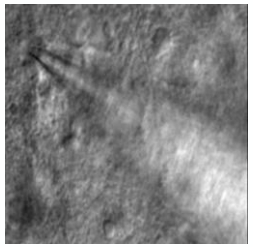


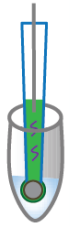
Biocytin imaging and morphological analysis

Patch clamp recording

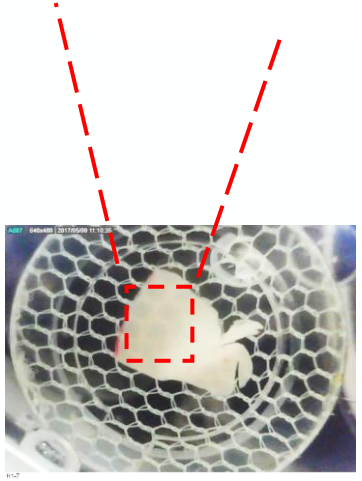








Cytosol/nuclear extraction



~300 um thick brain slices



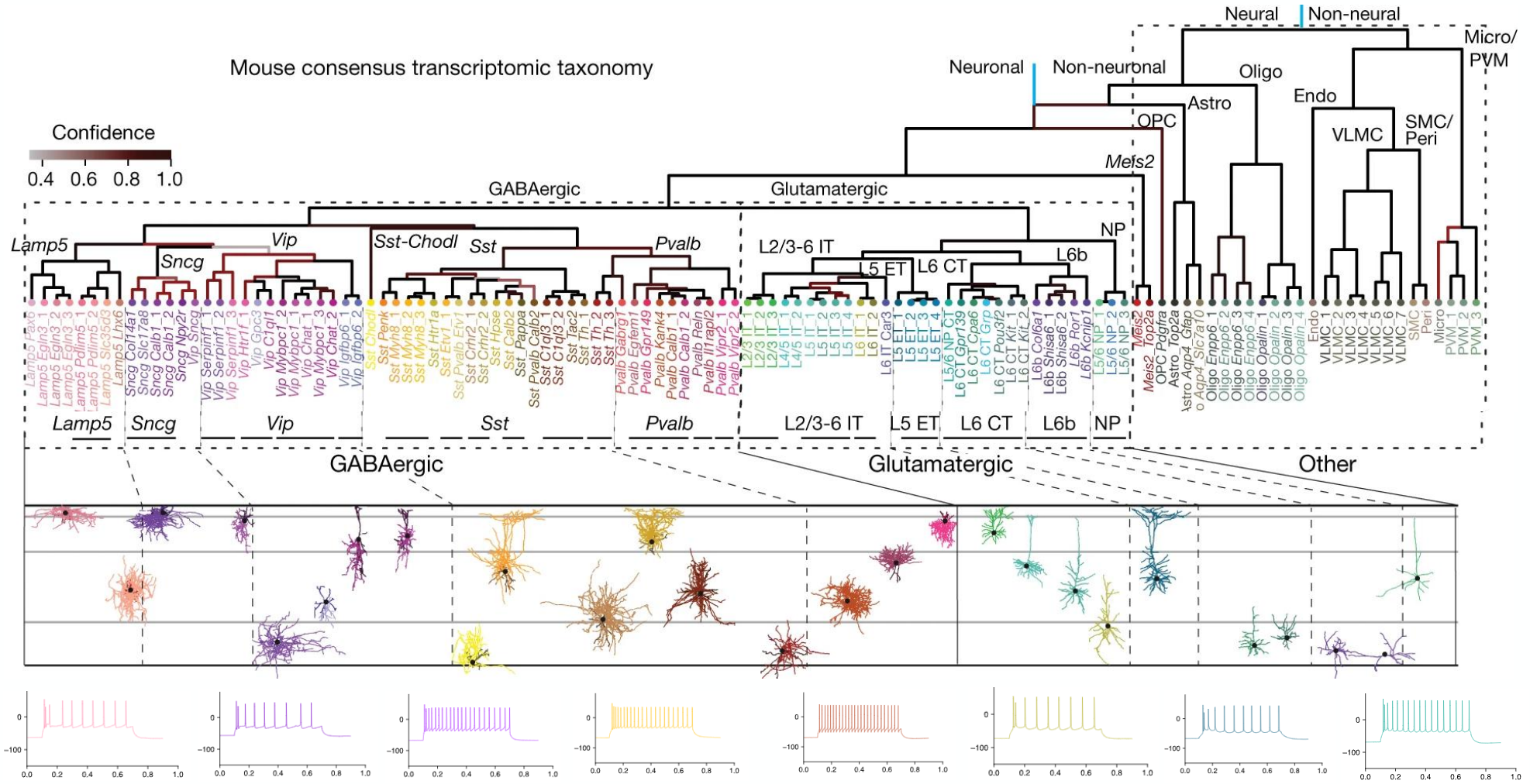
Ephys feature collection

patch-seq mouse motor cortex taxonomy

Gene Expression

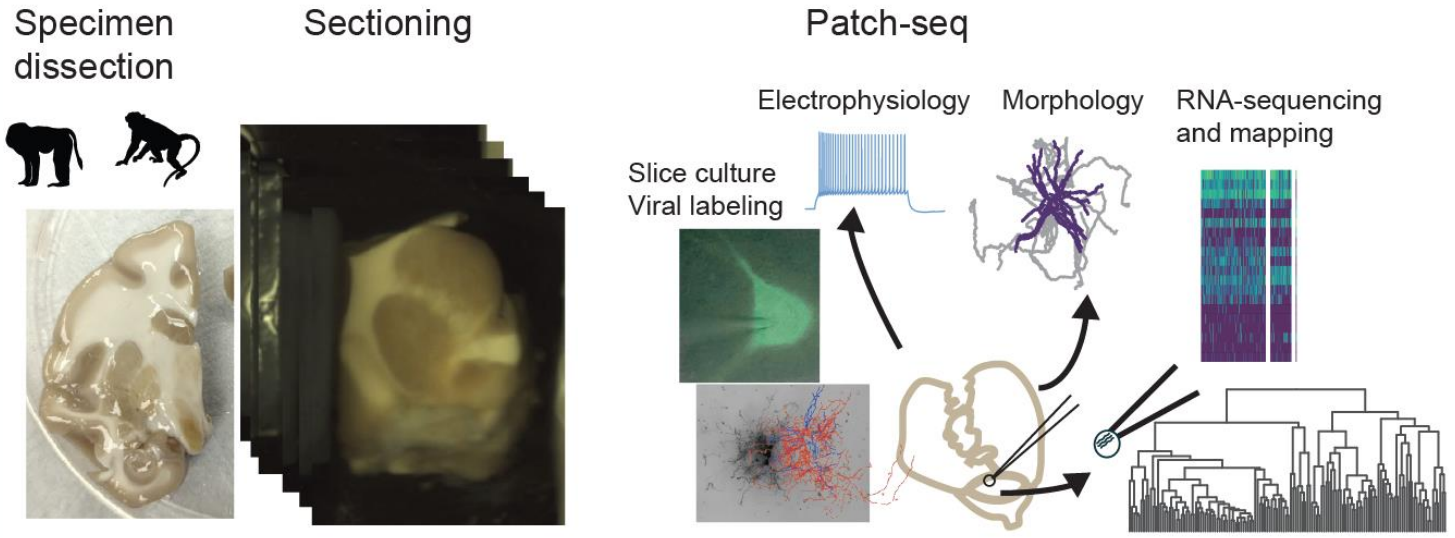
Morphology

Electrophysiology

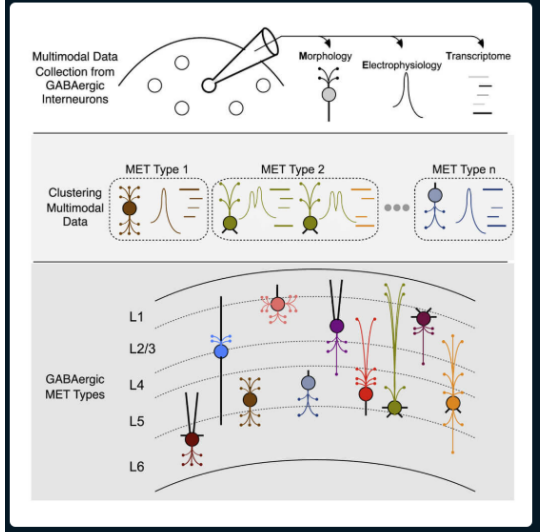


BICCN et al (2021)

NHP experimental preparation



viewing and accessing patch-seq data



Integrated Morphoelectric and Transcriptomic Classification of Cortical GABAergic Cells

This dataset includes 4,435 mouse inhibitory neurons from the visual cortex with transcriptomic, intrinsic physiological and, for a subset, morphological data. Neurons with all three modalities were assigned to one of 28 interneuron MET-types that have congruent morphological, electrophysiological, and transcriptomic properties.

Researchers can use the metadata file to find cells of a particular cell type of interest and then use the manifest file to download raw data files for those cells. Instructions and links are below. This video tutorial provides an overview of the data and walks through an example Jupyter notebook.

Explore the Data in the Mouse Patch-seq VIS Viewer

Gouwens, Sorensen et al. Cell 2020

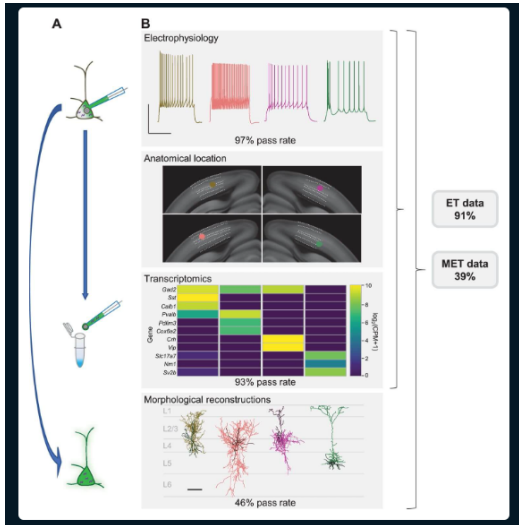
→ DOWNLOAD INSTRUCTIONS

→ DOWNLOAD CELL METADATA

→ DOWNLOAD RAW DATA MANIFEST FOR INDIVIDUAL CELLS

→ MORE DOWNLOADS

accessing patch-seq analysis tools

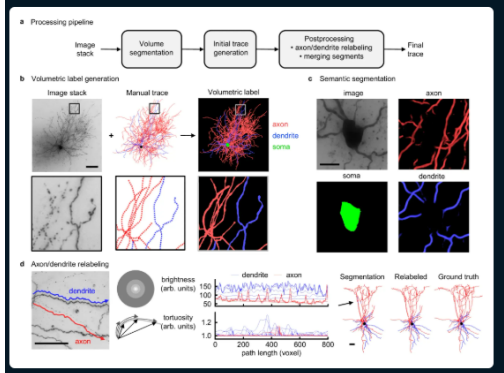


Scaled, high fidelity electrophysiological, morphological, and transcriptomic cell characterization

We have optimized the Patch-seq technique to efficiently collect high-quality electrophysiological, morphological and transcriptomic data. Here we describe this optimized technique as well as the publicly available tools that can be used to generate comparable data.

Patch-seq Analysis Tools (GitHub) | Community Forums

Lee, Budzillo et al. eLife 2021



High-throughput analysis of dendrite and axonal arbors reveals transcriptomic correlates of neuroanatomy

Introduces an automated pipeline for high-throughput reconstruction and analysis of dendritic and axonal arbors from brightfield-imaged neurons, enabling large-scale morphological characterization across transcriptomically defined cell types.

Analysis Code (GitHub) | Gliko, Mallory et al. Nature Communications 2024

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Troubleshooting Annotation with the BKP

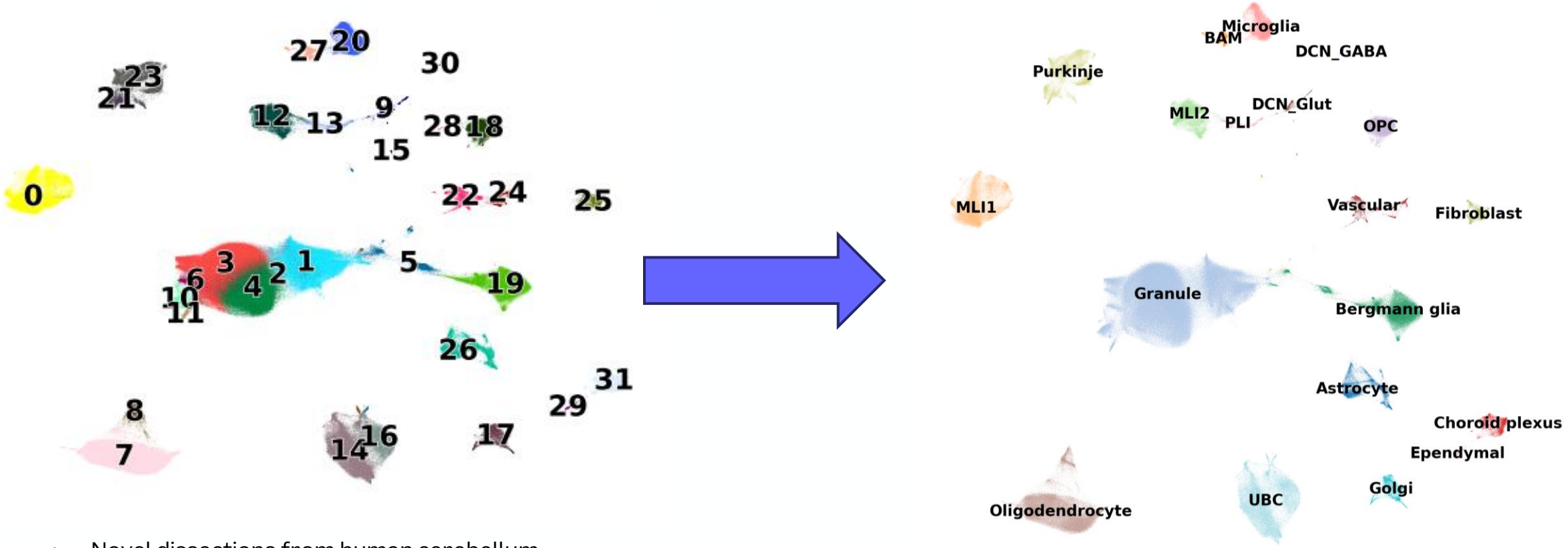
Leveraging the Brain Knowledge Platform when mapping fails...

Aaron Garcia, Scientist II

allen institute/

allen institute

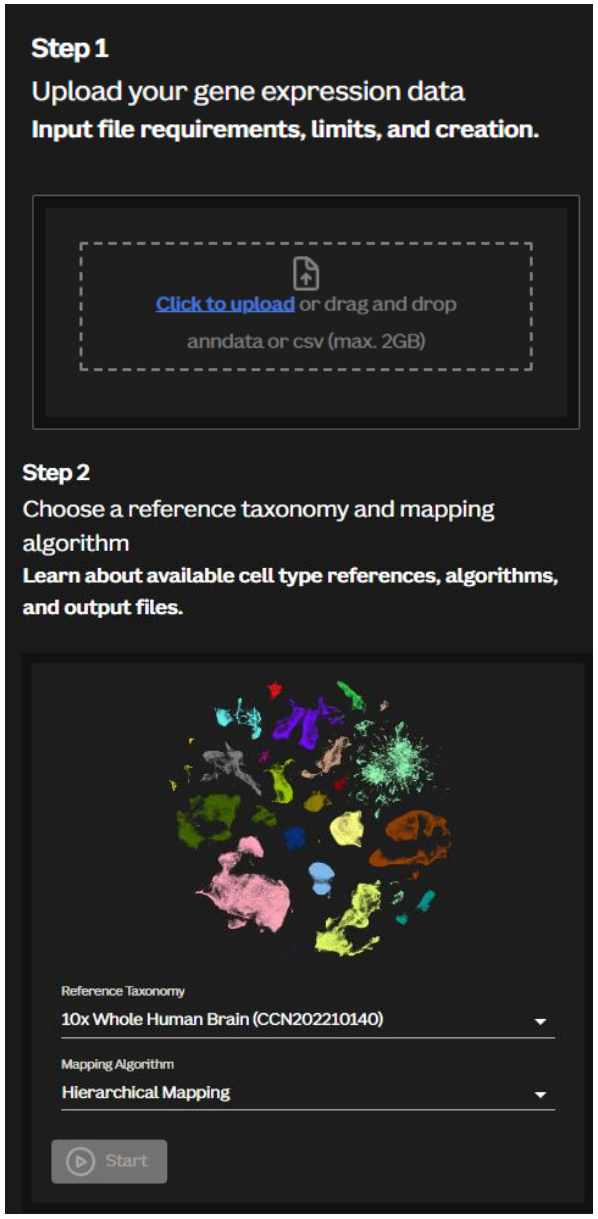
How do we go from novel data generation to comprehensive annotation?



- Novel dissections from human cerebellum
- Sequenced with 10X
- Pre-processed:
  - QC on n_genes, n_counts, doublets
  - Clustered

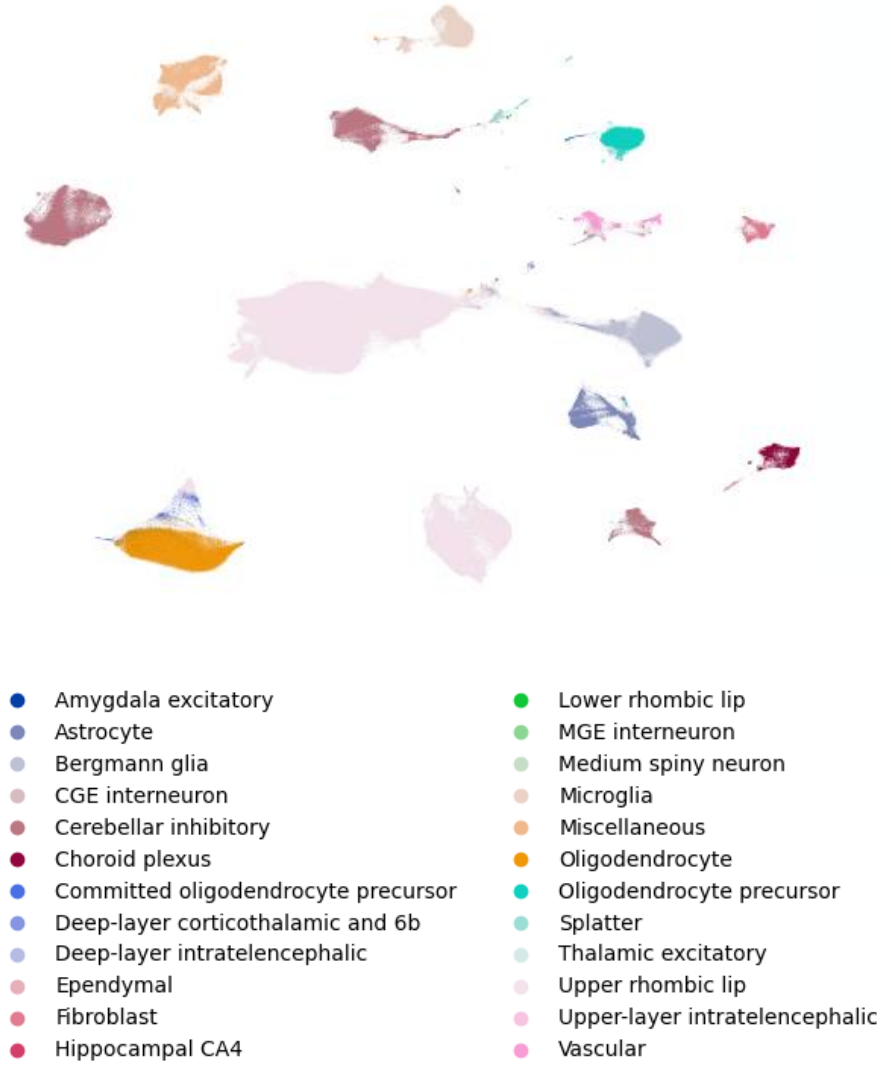
Usual first step: MapMyCells/Cell_type_mapper/

- MapMyCells
- CellTypeMapper Github Repository
- Why choose one over the other?
  - **MapMyCells:**
    - Pros: GUI-based, use our compute
    - Cons: Size, parameter limitations
  - **CellTypeMapper:**
    - Pros: No size restriction, more parameter tuning options
    - Cons: Requires python knowledge & your own compute
- Mapping algorithms:
  - **Correlation** – map directly to clusters, inherit overarching labels from best-match cluster
  - **Hierarchical** – deterministic walk down taxonomy tree
  - Learn more



MapMyCells mapping to whole human brain fails to capture cerebellar populations/

- Expected populations (i.e. Purkinje cells) not explicitly defined
- Lots of incorrect labels
- Mostly non-neuronal populations that receive a semi-accurate label
- **How do we resolve this?**

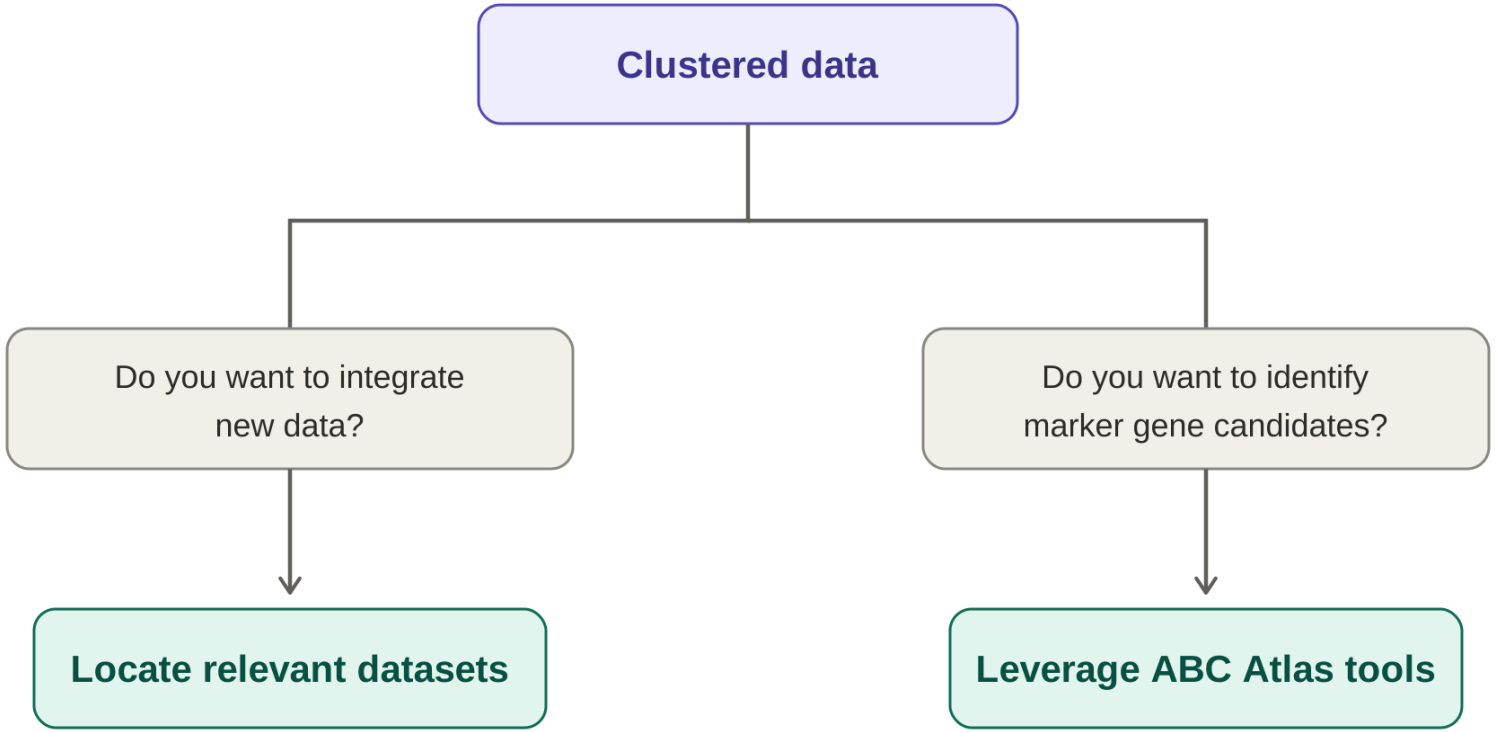


Where are the Purkinje cells?

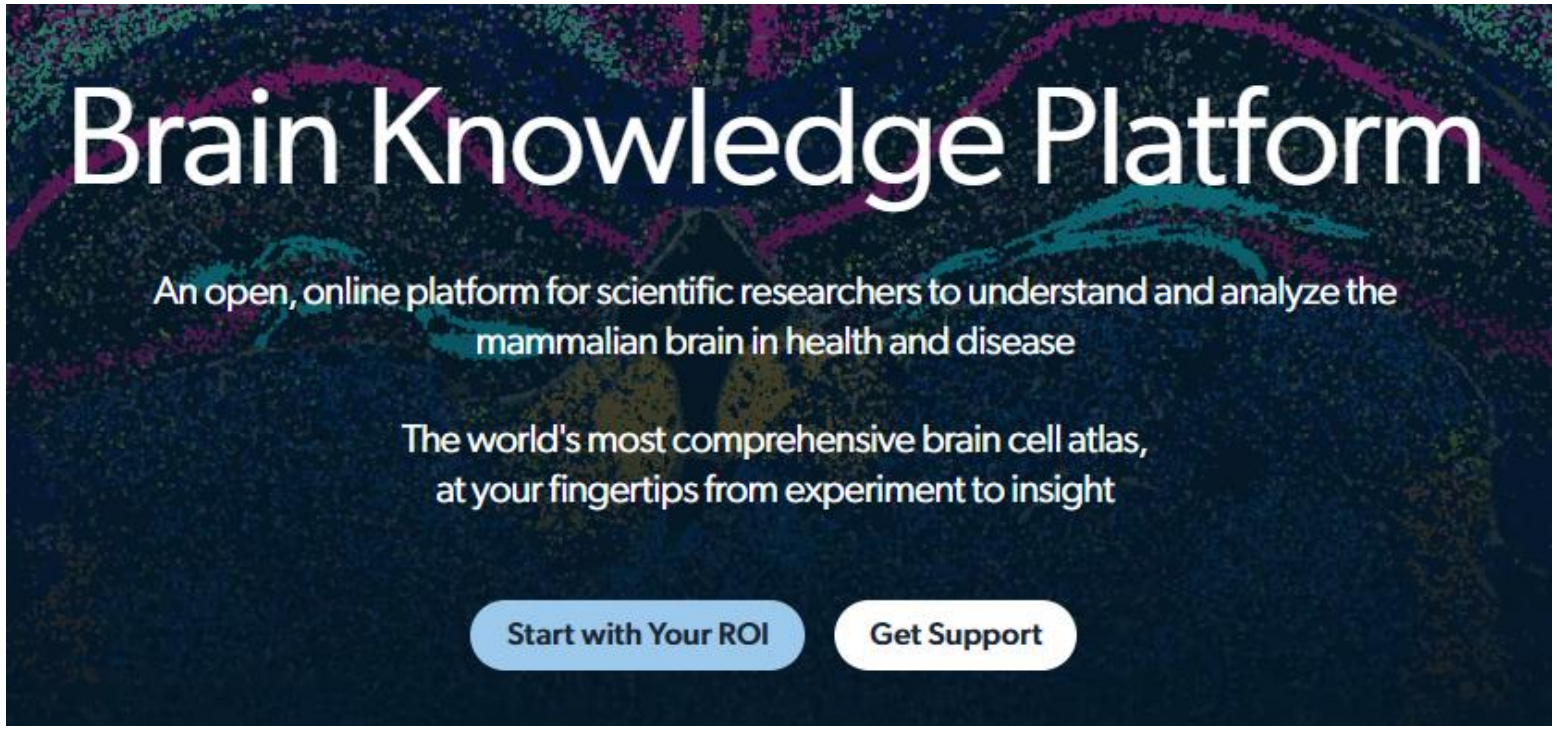


alleninstitute/

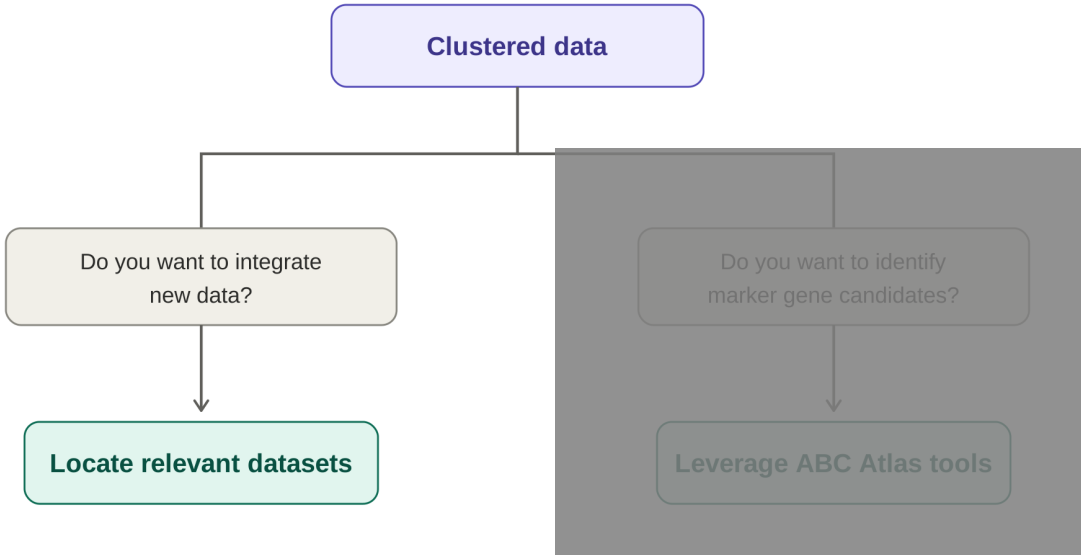
Workflow options/

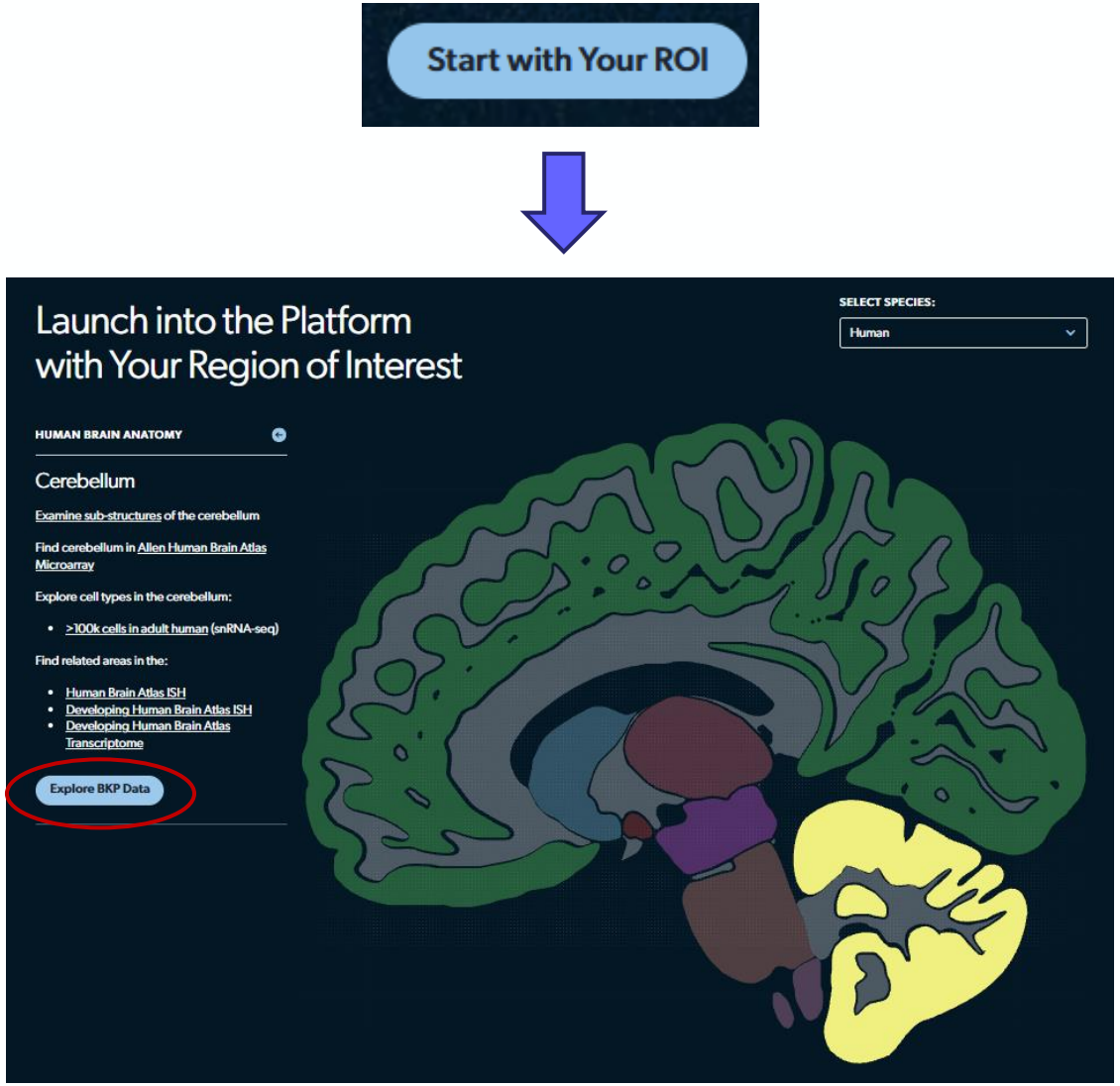


All roads lead to the Brain Knowledge Platform/

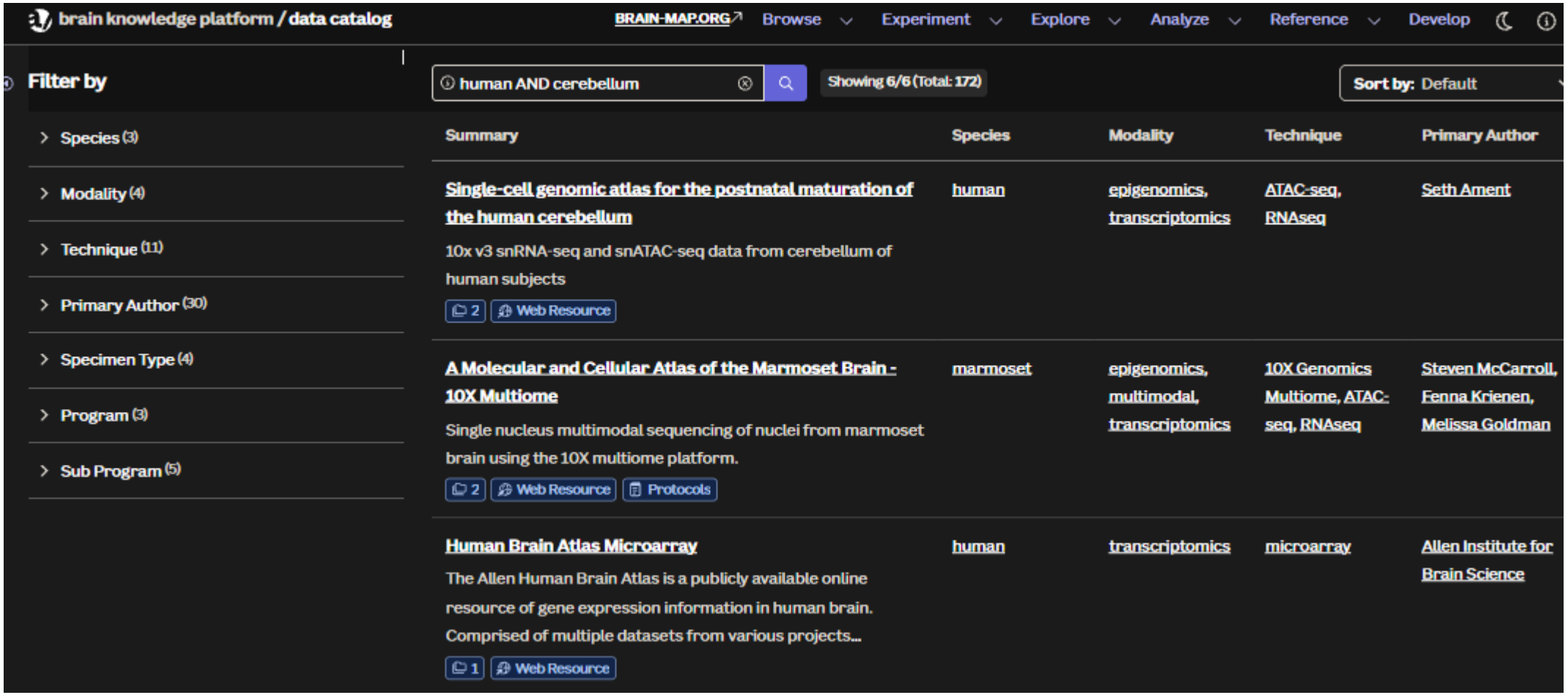


How do we locate relevant Allen Institute datasets?

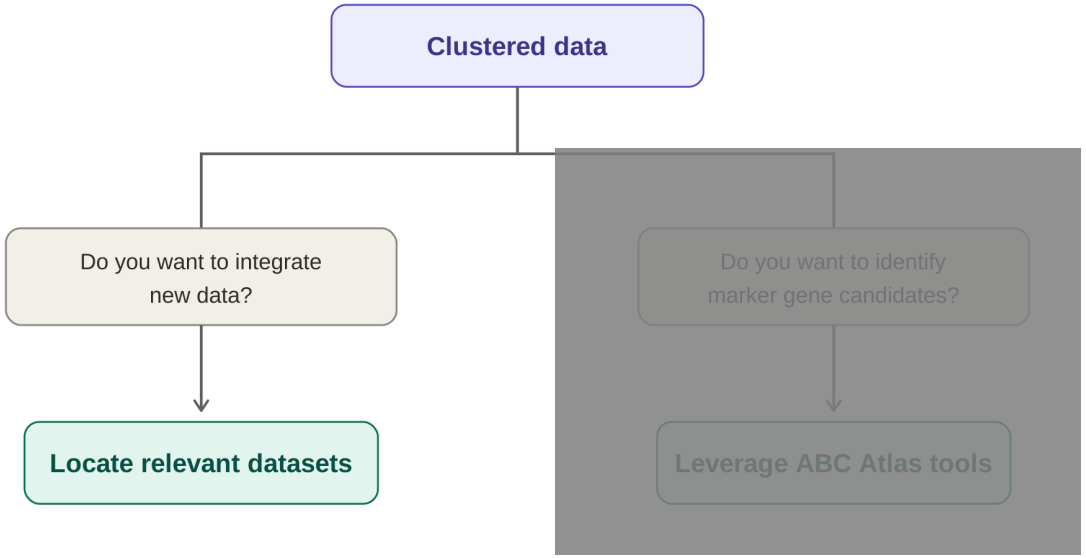


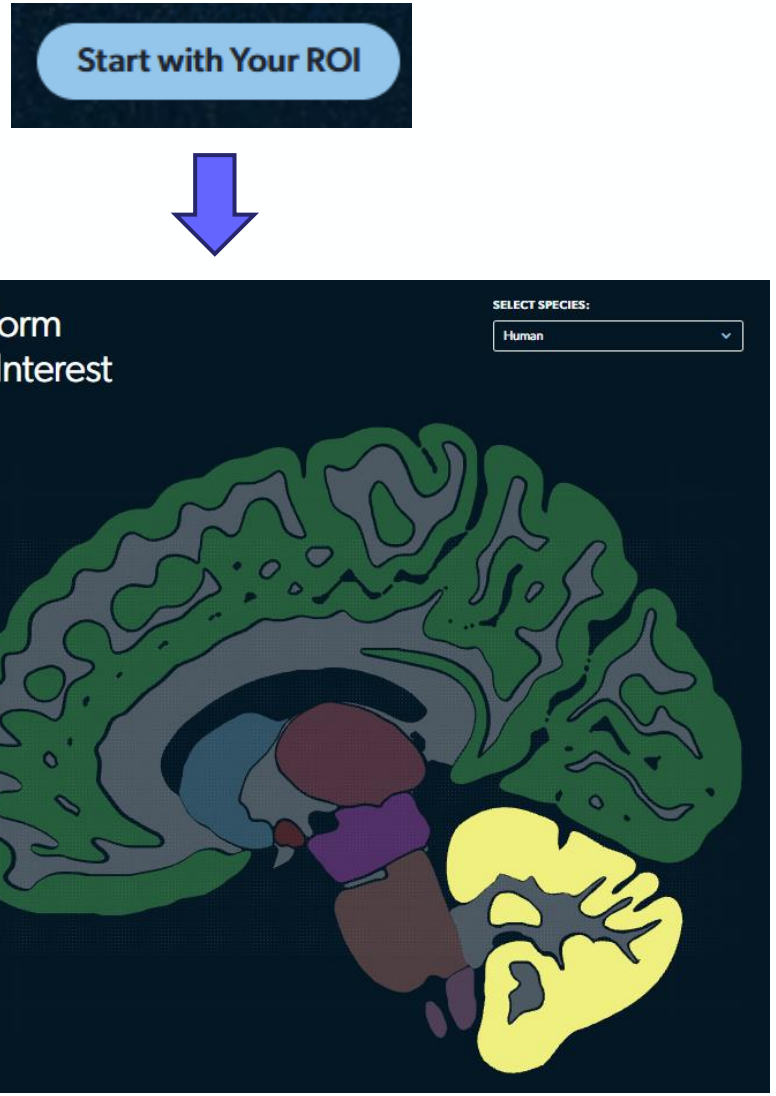


BKP data catalog contains multiple public-facing, multi-modal datasets for your integration needs

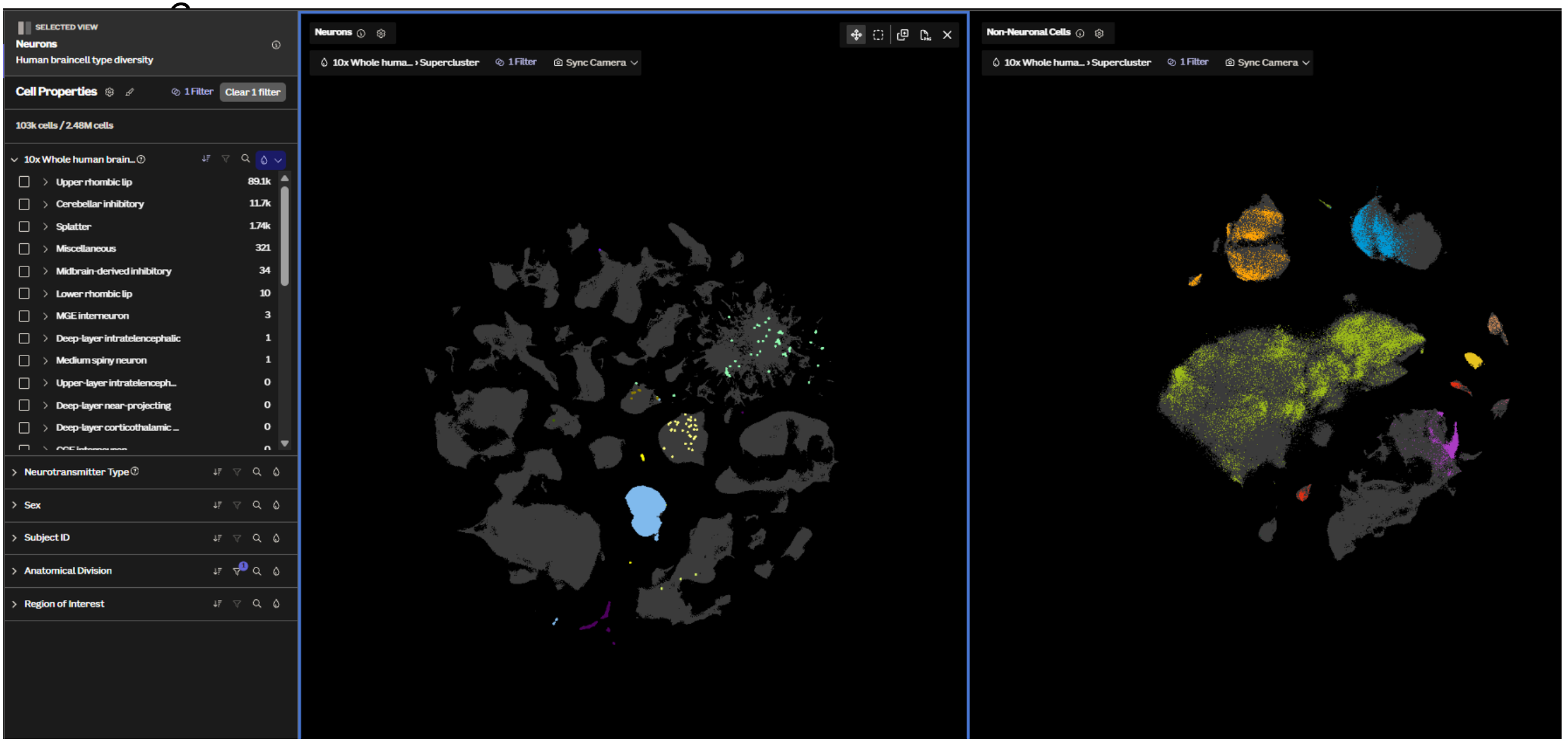


How do we visualize relevant Allen Institute datasets?

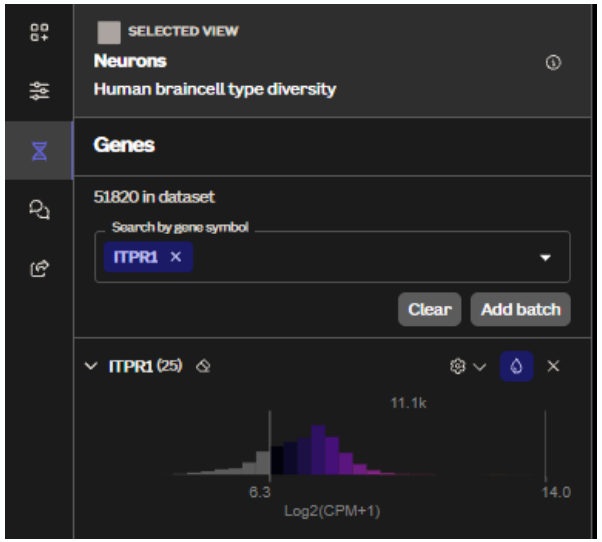


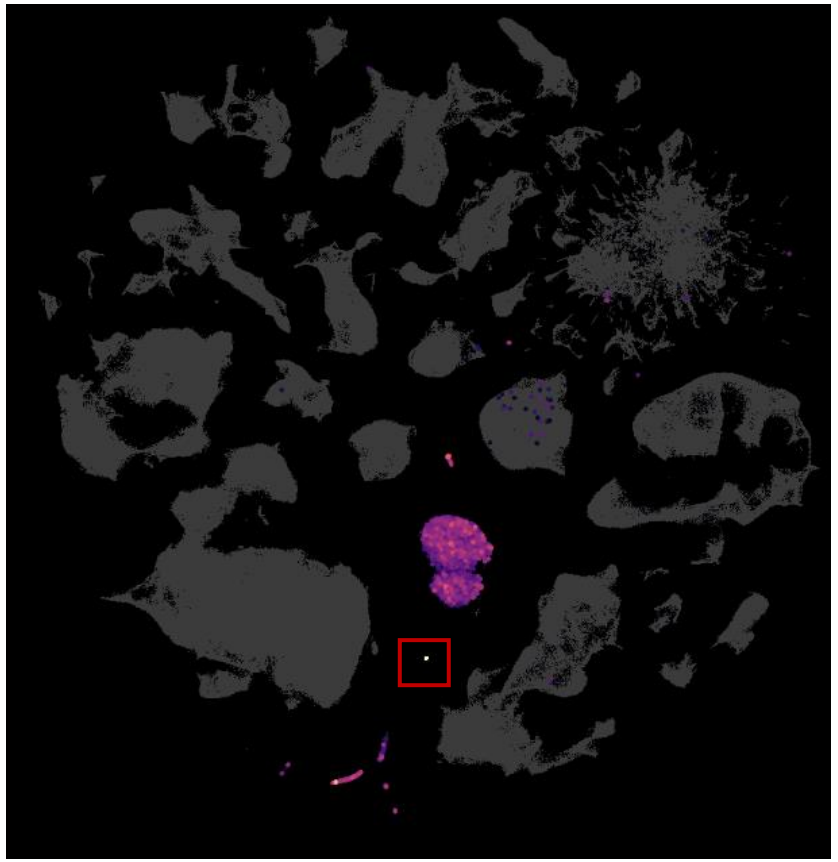


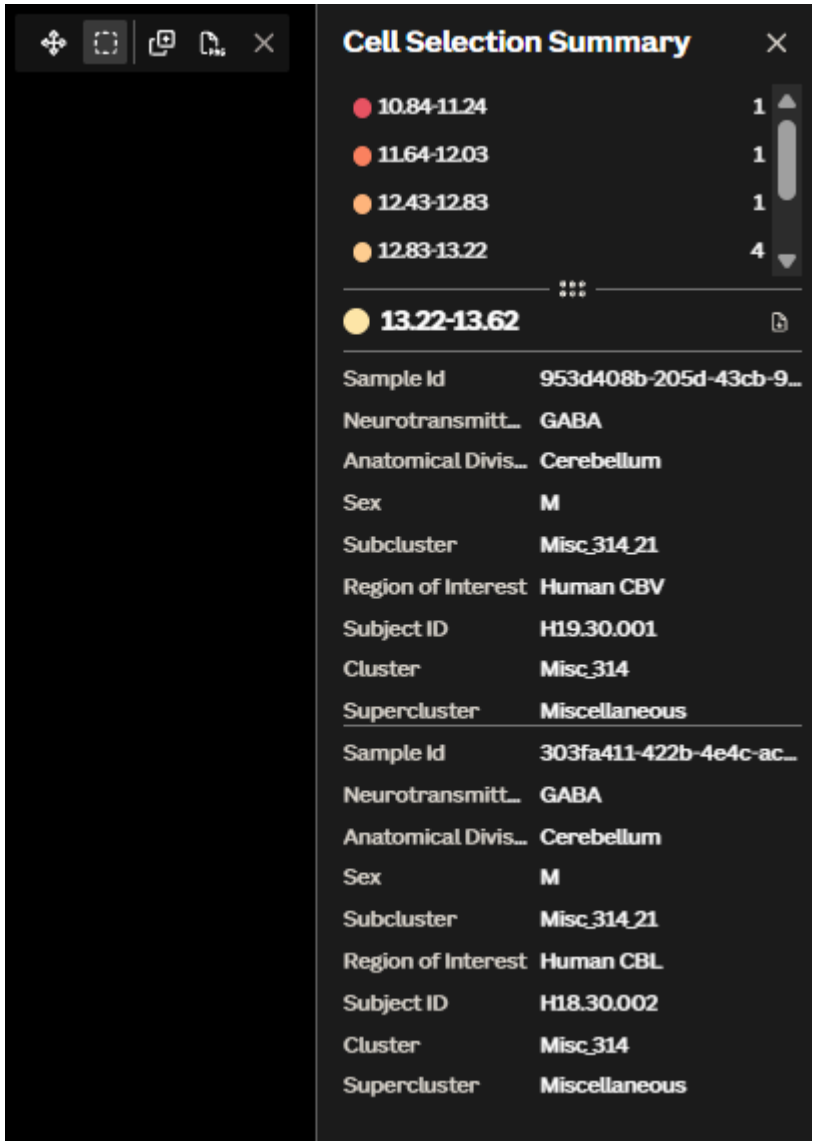
ABC Atlas can give us a filtered view of human cerebellar types/
unfortunately, our issue persists. But! Maybe we can investigate canonical



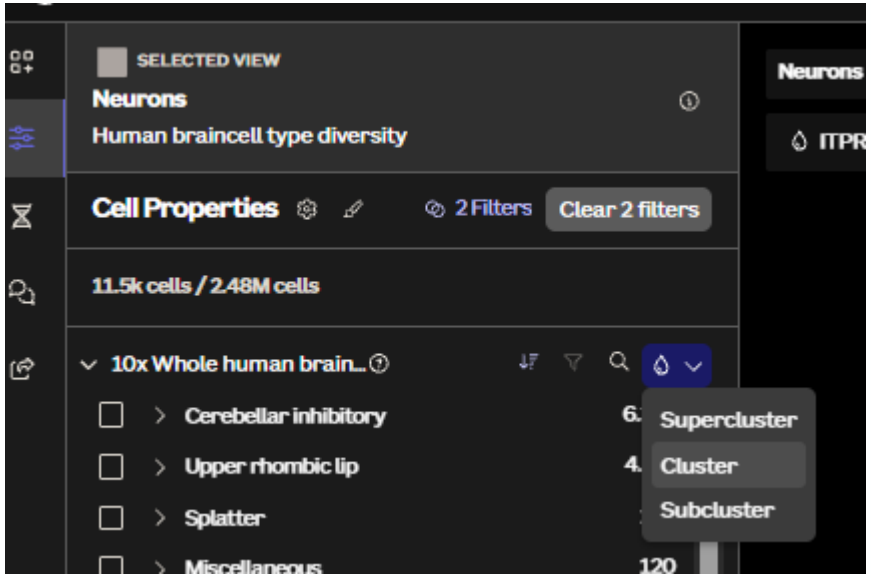
ITPR1 highlights a small cerebellar population/

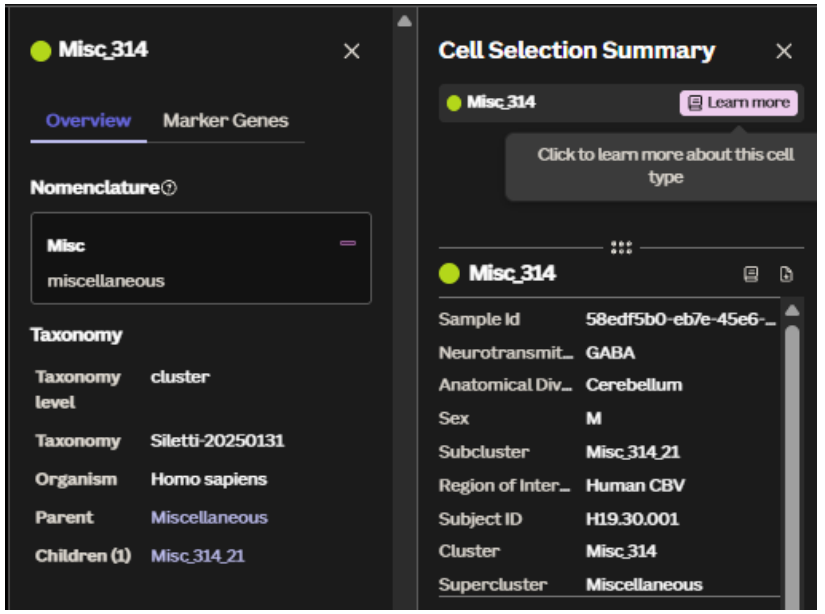


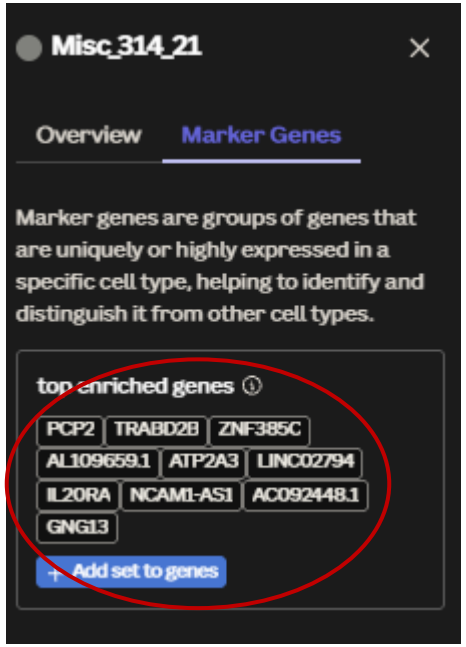




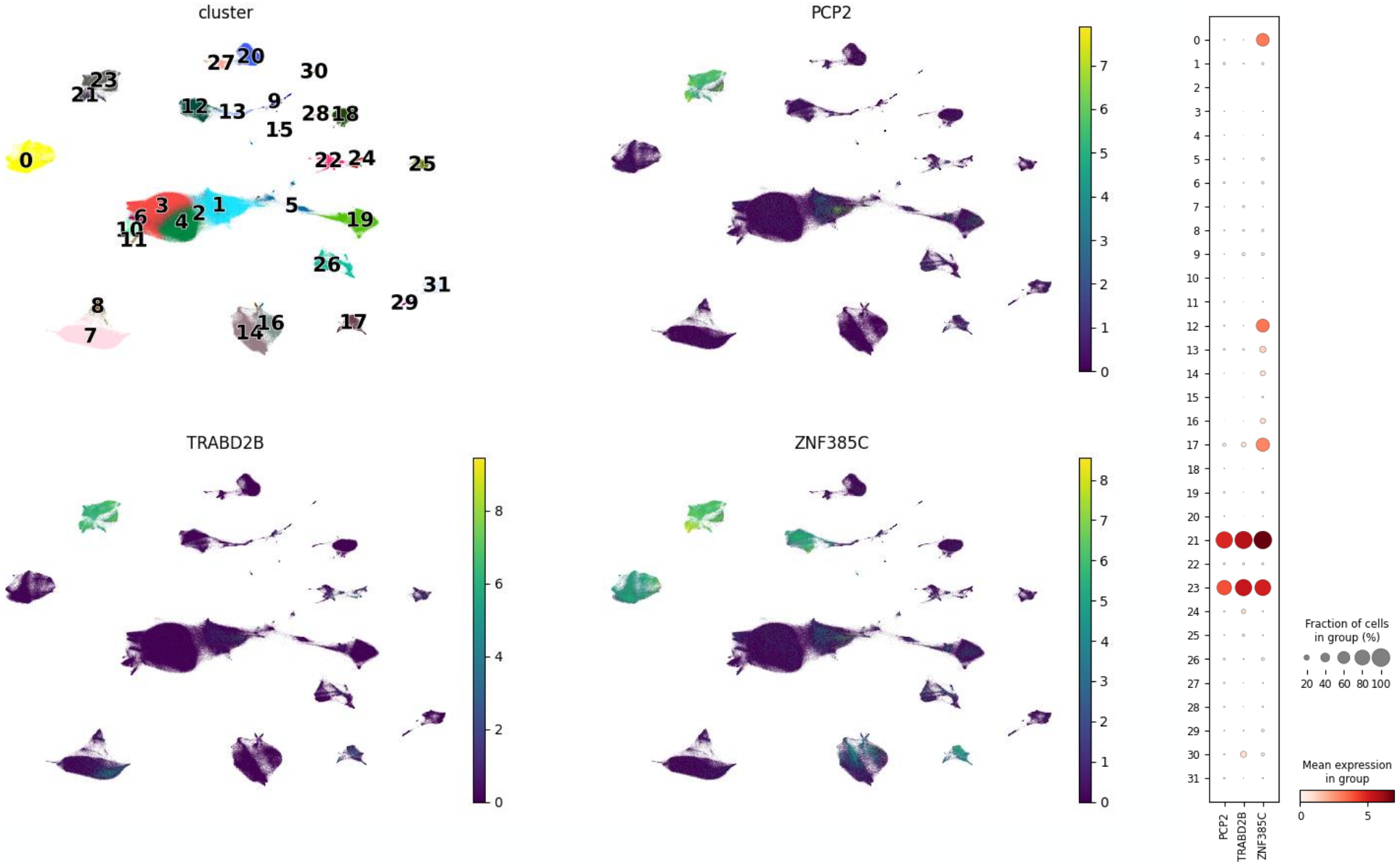
Flip back to color-by-taxonomy and “learn more” about our cluster-of-interest to find marker gene candidates/







Let’s see how these genes localize in our data/

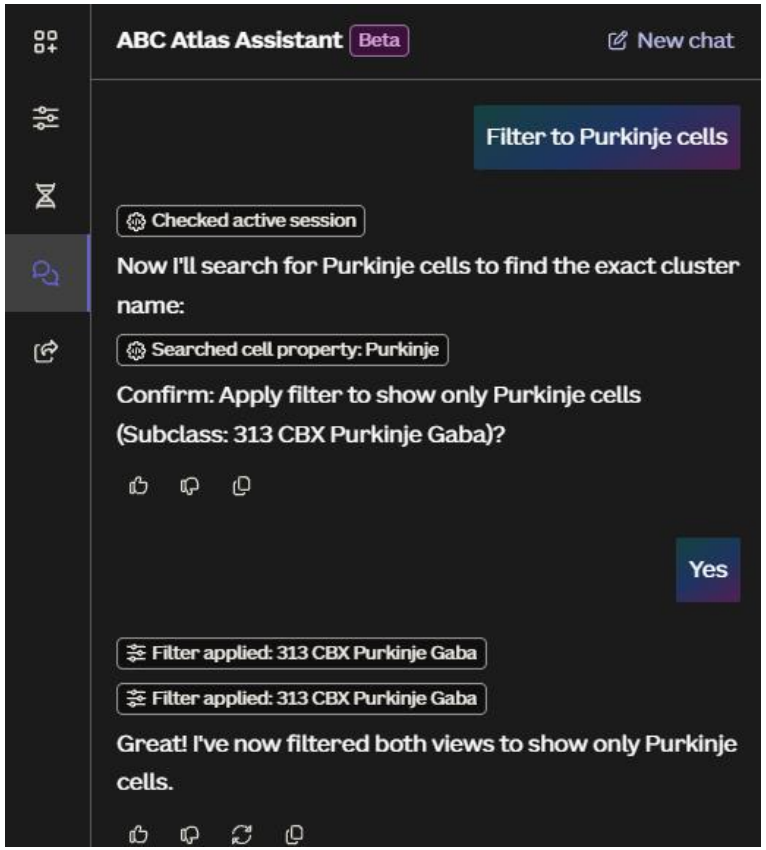


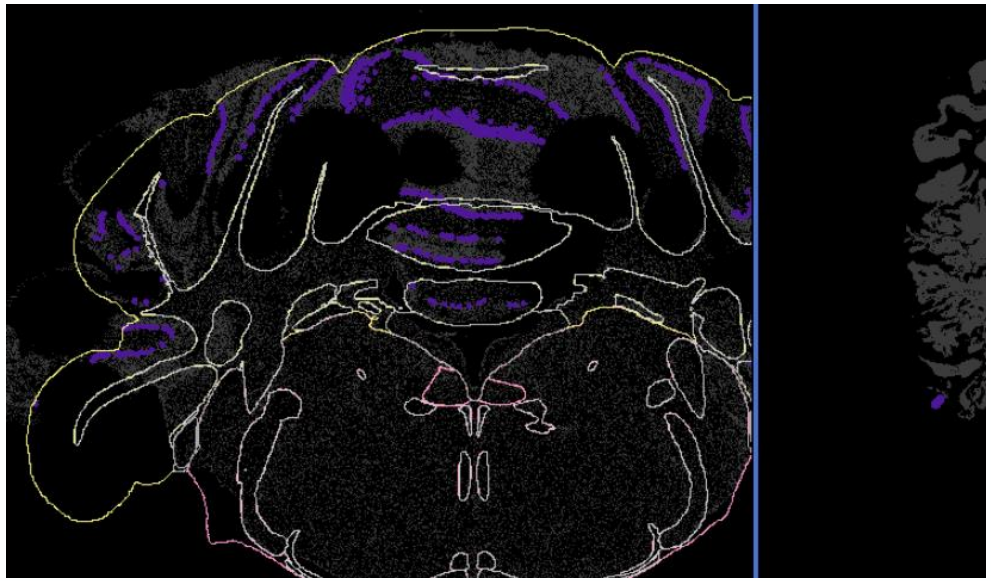
Using mouse markers and ABC Atlas Assistant to inform marker gene candidates/

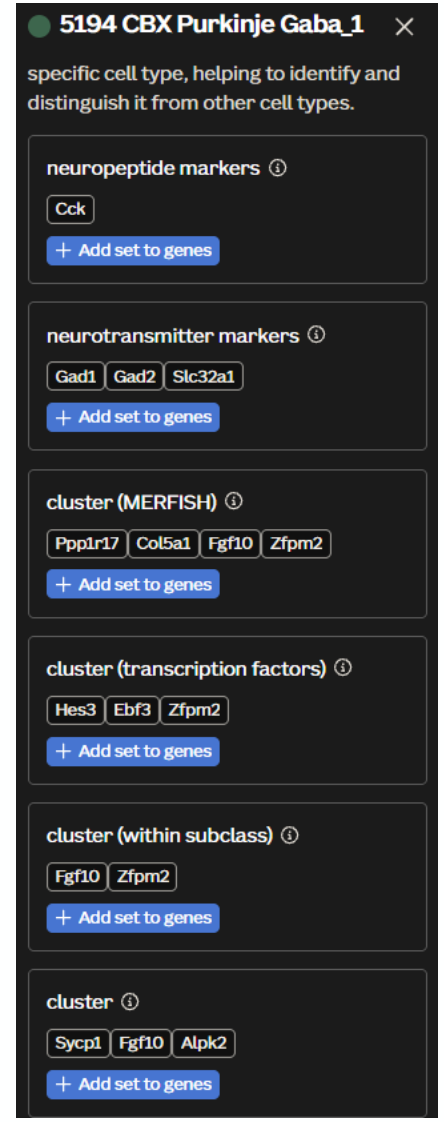




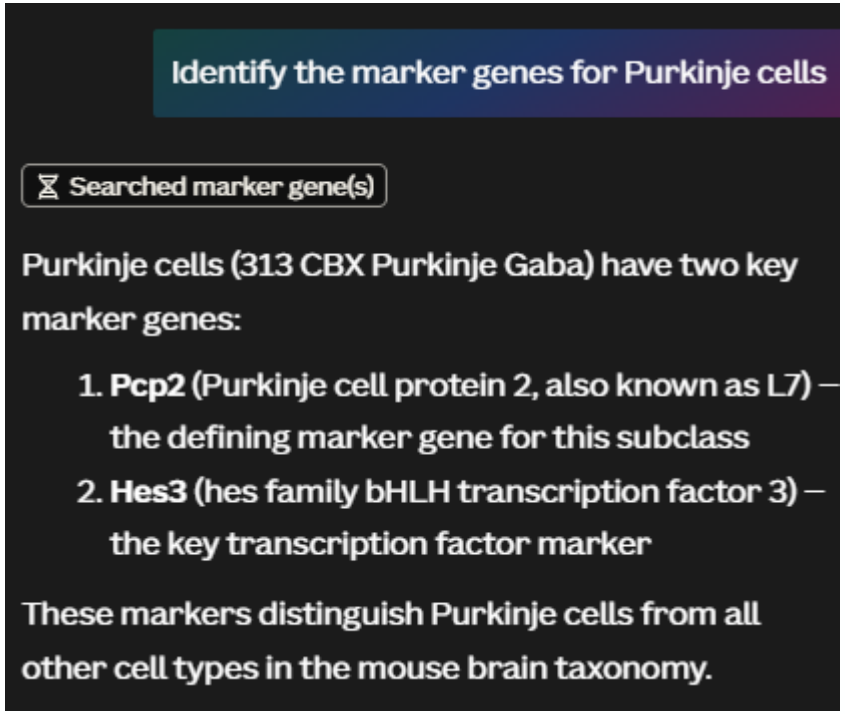
Using mouse markers and ABC Atlas Assistant to inform marker gene candidates/Similar end step to human data





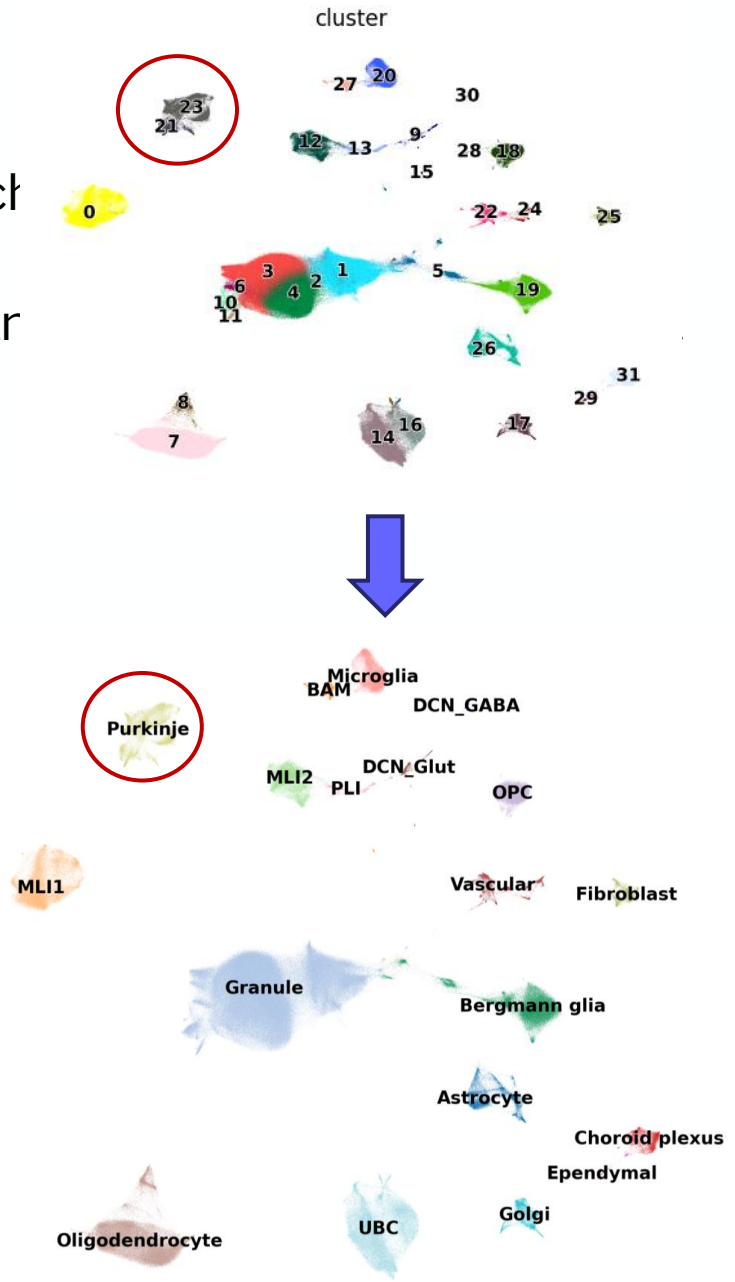


Using mouse markers and ABC Atlas Assistant to inform marker gene candidates/Use assistant to aggregate gene names



Summary/

- Mapping to existing taxonomies may require follow-up to acl resolution.
- BKP provides a comprehensive platform for locating relevar analyses.
  - Locate additional dataset
  - Use visualization tools to identify candidate marker genes
- Leveraging ABC Atlas, we can:
  - Use canonical genes to locate putative cells from mapping datasets
  - Use mouse ABC atlas to identify candidate marker genes to apply to o
    - Note: This may require validation that genes are orthologous
- Applied to our novel cerebellum data, we can ID Purkinje neurons and expa
- Stay tuned for updates to primate whole brain efforts with HMBA!



Please let us know if you identify potential taxonomy updates!

Neocortex	Rebecca Hodge
Medial temporal lobe	Aaron Garcia, Yuanyuan Fu
Basal ganglia	Nelson Johansen, Yuanyuan Fu
Thalamus	Brian Long, Meghan Turner
Hypothalamus	Fenna Krienen, Stephanie Seeman
Midbrain, Medulla, Pons	Zizhen Yao, Cindy van Velthoven, Hongkui Zeng
Cerebellum	Rebecca Hodge
Spinal cord	Matthew Schmitz, Nelson Johansen
Non-neuronal WB	Yuanyuan Fu

ALLEN BRAIN MAP
COMMUNITY FORUM

Additional/External Python packages used in this tutorial

- Standard Python scRNAseq package: scanpy
  - Scanpy.pl.embedding() – flexible UMAP plotting
  - Scanpy.pl.dotplot() – Gene expression dot plots
- Quality Control - Sciduck
- Embedding space generation
  - scVI
  - scanpy – PCA>Neighbors graph>UMAP


alleninstitute/
thank you/

agenda/afternoon **– breakout rooms**

Session 1: connecting multimodal data to brain cell types

Room 1255-1265

1:10-1:50 | defining and using common coordinate frameworks (reference atlases, mesoscale)

1:50-2:00 | patch-seq overview and tools for analysis

2:00-2:20 | Troubleshooting Annotation with the BKP

2:20-2:30 | break

2:30-2:40 | ABC Atlas demo & external use case examples

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ABC Atlas demo & external use case examples

allen institute/

community use of BKP tools and data/

Title: Evolution of Cajal-Retzius Cells in Vertebrates from an Ancient Class of Tp73+ Neurons

Authors: Elias Gumnit[1], Giacomo Gattoni[1], Jamie Woych[1], Astrid Deryckere[1#], Prabesh Bhattarai[2,3], J Andrew Gillis[4], Caghan Kizil[2,3], Maria Antonietta Tosches[1*]

Next, we investigated whether the molecular and spatial heterogeneity of salamander *Tp73*+ neurons has counterparts in mammals. We isolated mouse *Tp73*+ neurons from the Allen Brain Cell (ABC) atlas (*34*) and compared them to the salamander *Tp73*+ neuron dataset. The

A unified analysis of atlas single-cell data

Hao Chen,[1,2] Nam D. Nguyen,[1] Matthew Ruffalo,[1] and Ziv Bar-Joseph[1,3]

To further test the ability of GIANT on integrating data from different modalities, we collected single-cell mouse brain data from the Allen Brain Cell Atlas (Yao et al. 2023), which contains 2,341,350 cells from scRNA-seq and 3,938,808 cells from MERFISH spatial transcriptomics (Chen et al. 2015). Both modali-

Morphine regulates astrocyte transcriptional dynamics in the ventral tegmental area by stimulation of glucocorticoid signaling

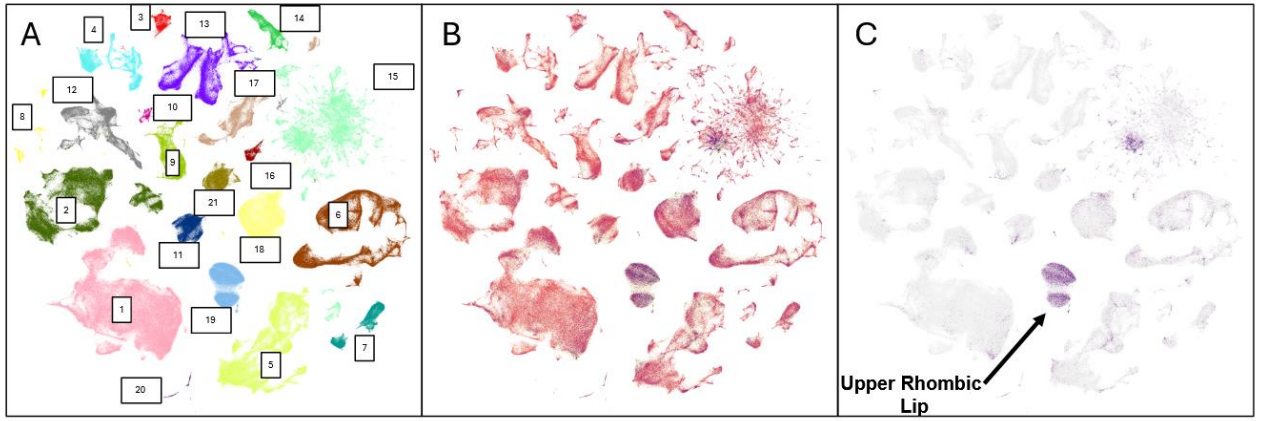
Jennifer J. Tuscher[1,3]*, Angela Cleere[1,3], Robert A. Phillips III[1,4], Catherine E. Newman[1], Guy Twa[1], Nathaniel J. Robinson[1], Lara Ianov[1], Robert E. Sorge[2], & Jeremy J. Day[1]*

clusters using lisi v1.0 in R[83]. To refine resolution of VTA neuronal subtypes and assign cells to classes, we followed the Allen Institute's Whole Mouse Brain taxonomy using the MapMyCells web interface[84] (RRID:SCR_024672). After removing additional low-

Review

Using the Allen Brain Cell Atlas of the Human Brain to Gain Insights into C-Terminal-Binding Protein 1 (CtBP1)'s Potential Function

Suhjin Lee [1] and Uthayashanker R. Ezekiel [2,*]



Use cases for ABC Atlas

“Are genes X and Y co-expressed in the same cell type?”
“Where is gene X expressed in the brain?”
“Is gene X expressed in a specific cell type?”
“Where is cell type X in the brain?”
“What is the spatial relationship between cell type X and cell type Y?”

However,

- MERFISH data in ABC Atlas is based on cell segmentation; cell-to-cell or synaptic contacts are not shown.
  - “Does cell type X synapse onto cell type Y?” cannot be answered
- MERFISH data in ABC Atlas shows centroids of cells, not cell shape or size
  - “Is the soma of cell type X larger than the soma of cell type Y?” cannot be answered

ABC atlas data/ experimental design

New Results

Follow this preprint

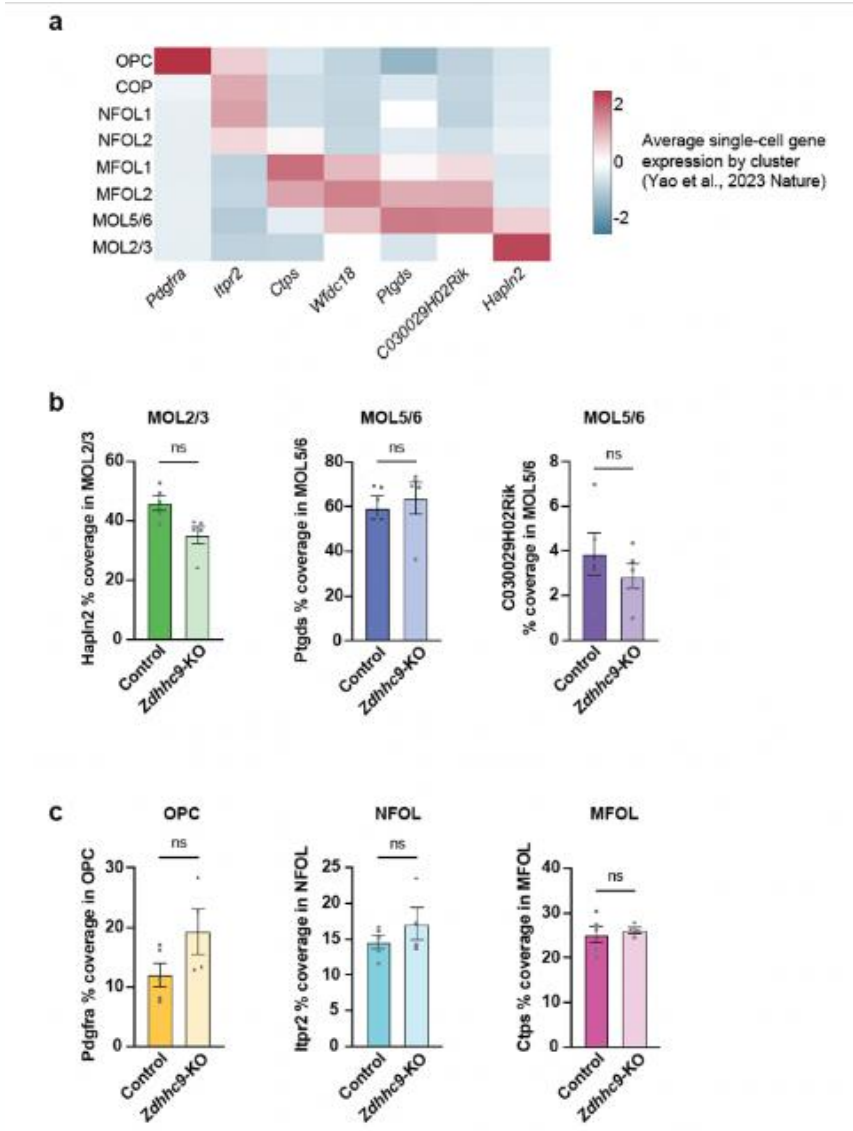
The X-Linked Intellectual Disability gene, *ZDHHC9*, is important for oligodendrocyte subtype determination and myelination

Experimental design: designing FISH probes to label oligo subtypes based on marker genes found in ABC Atlas cell types

Altered oligodendrocyte subtype proportions in *Zdhhc9*-KO mice

The changes in gene expression identified with RNAseq predict alterations in proportion of oligodendrocyte lineage cells in *Zdhhc9*-KO mice (Figure 1d). We therefore designed fluorescent *in situ* hybridization (FISH RNAscope) probes for the dominant MOL subtypes in the corpus callosum, guided by expression data from previously published scRNAseq and FISH studies(15, 28, 32)(Figure S4a). We chose marker genes with highly enriched expression in specific oligodendrocyte lineage cells, including *Hapln2* (MOL2/3 marker), *Ptgds* (MOL5/6 marker), and *C030029H02Rik* (MOL5/6 marker, Figure S4a). Importantly, within each cell type identified we

Hollman et al., 2024 (preprint)



ABC atlas data/ data analysis

Home > The EMBO Journal > Article

Rhythmic astrocytic GABA production synchronizes neuronal circadian timekeeping in the suprachiasmatic nucleus

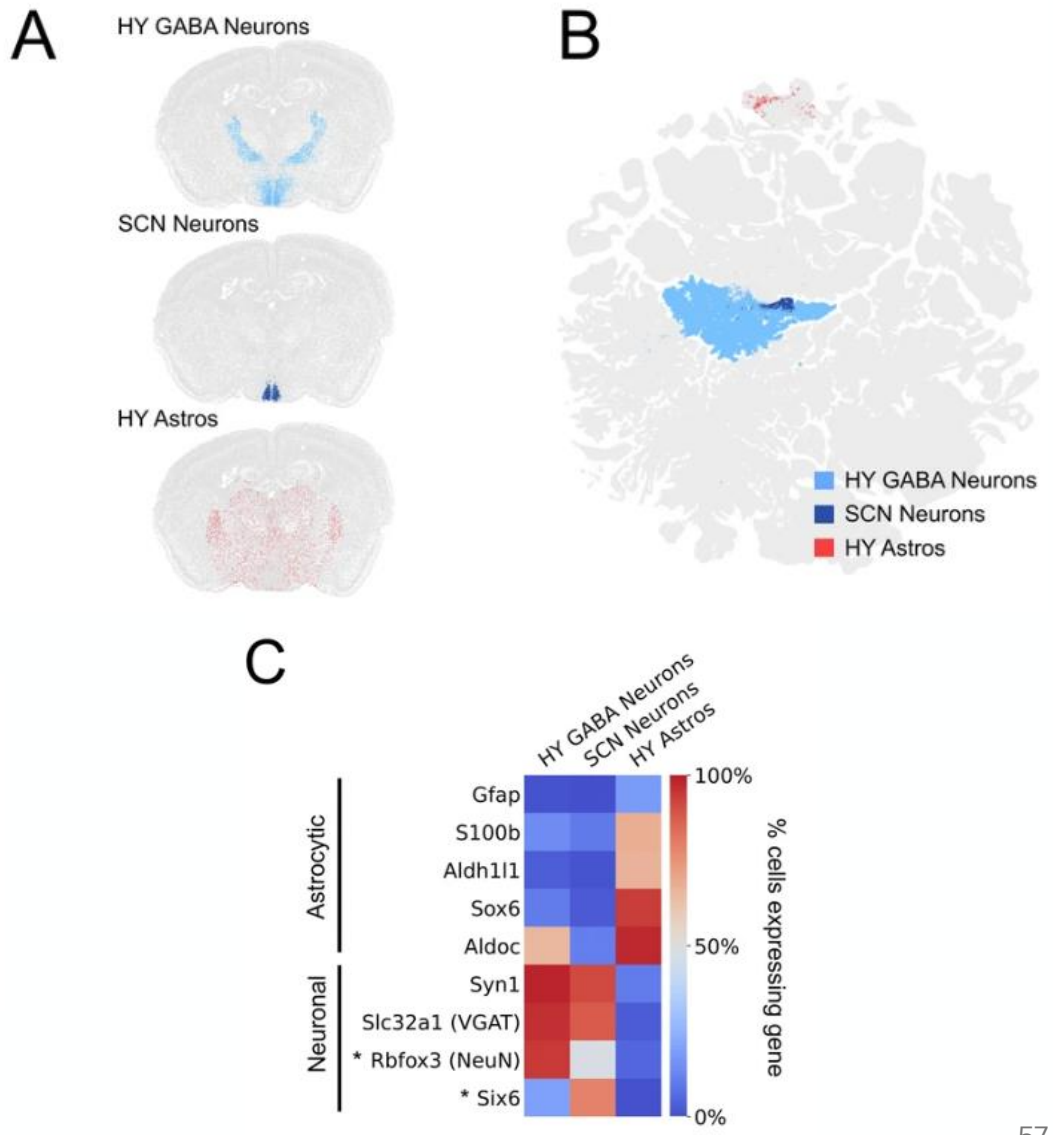
Article | Open access | Published: 02 December 2024

Volume 44, article number 2 (2025) Cite this article

Data analysis: examining synthesis pathways in astrocytes and neurons from ABC Atlas

To examine whether these astrocytic GABA synthesis pathways are expressed in SCN astrocytes, we analyzed publicly available data from a recent large single-cell RNA sequencing dataset of the mouse brain, consisting of a total of ~2.3 million cells, by the Allen Brain Institute (Yao et al, 2023). We compared gene expression of relevant biosynthetic pathways in hypothalamic astrocytes, SCN neurons and hypothalamic GABAergic neurons (Fig. 4A,B). Hypothalamic GABAergic neurons, SCN neurons and

Ness et al., 2024



ABC atlas data/ novel tool generation

New Results

Follow this preprint

3D multimodal histological atlas and coordinate framework for the mouse brain and head

Novel tool: mouse brain and head atlas combining MRI & CT imaging with histology data

Tward et al., 2025 (preprint)

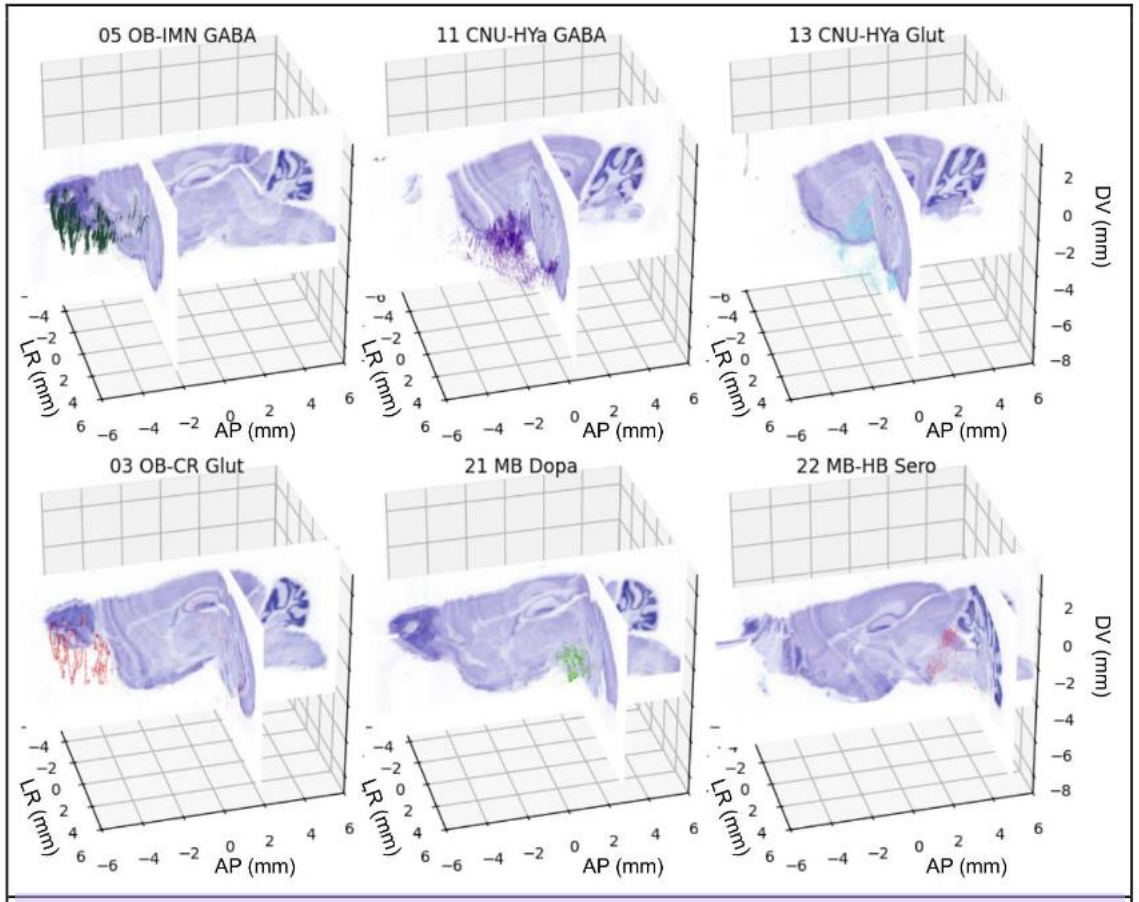


Figure 5. Spatial distributions of 6 selected cell types (see plot titles) from the ABC dataset, with coordinates mapped into our coordinate system. A 3D reconstructed Nissl reference brain dataset is shown in our coordinate system. Coronal and sagittal sections are chosen such that that 90% of cells appear in front of the slices. The visualization illustrates the intact brainstem present in the Nissl reference volume. Spatial distributions of a larger set of cell types are shown in **Extended Data Figure 6**.

ABC Atlas demo

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feedback survey/ please fill out!

www.surveymonkey.com/r/7VMVBKR


alleninstitute/
thank you/

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